3.2 At what distance from a 100-W light bulb is the radiant flux equal to the solar irradiance?

3.3 The parallax angle for Sirius is $0.379''$.

(a) Find the distance to Sirius in units of (i) parsecs; (ii) light-years; (iii) AU; (iv) m.

(b) Determine the distance modulus for Sirius.

3.4 Using the information in Example 3.6.1 and Problem 3.3, determine the absolute bolometric magnitude of Sirius and compare it with that of the Sun. What is the ratio of Sirius's luminosity to that of the Sun?

3.5 **(a)** The Hipparcos Space Astrometry Mission was able to measure parallax angles down to nearly $0.001''$. To get a sense of that level of resolution, how far from a dime would you need to be to observe it subtending an angle of $0.001''$? (The diameter of a dime is approximately 1.9 cm.)

(b) Assume that grass grows at the rate of 5 cm per week.

i. How much does grass grow in one second?

ii. How far from the grass would you need to be to see it grow at an angular rate of $0.000004''$ (4 microarcseconds) per second? Four microarcseconds is the estimated angular resolution of SIM, NASA's planned astrometric mission; see page 59.

3.6 Derive the relation

$$m = M_{\text{Sun}} - 2.5 \log_{10} \left(\frac{F}{F_{10,\odot}} \right).$$

3.7 A 1.2×10^4 kg spacecraft is launched from Earth and is to be accelerated radially away from the Sun using a circular solar sail. The initial acceleration of the spacecraft is to be $1g$. Assuming a flat sail, determine the radius of the sail if it is

(a) black, so it absorbs the Sun's light.

(b) shiny, so it reflects the Sun's light.

Hint: The spacecraft, like Earth, is orbiting the Sun. Should you include the Sun's gravity in your calculation?

3.8 The average person has 1.4 m^2 of skin at a skin temperature of roughly 306 K (92°F). Consider the average person to be an ideal radiator standing in a room at a temperature of 293 K (68°F).

(a) Calculate the energy per second radiated by the average person in the form of blackbody radiation. Express your answer in watts.

(b) Determine the peak wavelength λ_{max} of the blackbody radiation emitted by the average person. In what region of the electromagnetic spectrum is this wavelength found?

(c) A blackbody also absorbs energy from its environment, in this case from the 293-K room. The equation describing the absorption is the same as the equation describing the emission of blackbody radiation, Eq. (3.16). Calculate the energy per second absorbed by the average person, expressed in watts.

(d) Calculate the net energy per second lost by the average person via blackbody radiation.

3.9 Consider a model of the star Dschubba (δ Sco), the center star in the head of the constellation Scorpius. Assume that Dschubba is a spherical blackbody with a surface temperature of 28,000 K and a radius of 5.16×10^9 m. Let this model star be located at a distance of 123 pc from Earth. Determine the following for the star:

(a) Luminosity.

(b) Absolute bolometric magnitude.

(c) Apparent bolometric magnitude.

(d) Distance modulus.

(e) Radiant flux at the star's surface.

(f) Radiant flux at Earth's surface (compare this with the solar irradiance).

(g) Peak wavelength $\lambda_{\max}$.

3.10 **(a)** Show that the Rayleigh–Jeans law (Eq. 3.20) is an approximation of the Planck function B_λ in the limit of $\lambda \gg hc/kT$. (The first-order expansion $e^x \approx 1 + x$ for $x \ll 1$ will be useful.) Notice that Planck's constant is not present in your answer. The Rayleigh–Jeans law is a *classical* result, so the "ultraviolet catastrophe" at short wavelengths, produced by the λ^4 in the denominator, cannot be avoided.

(b) Plot the Planck function B_λ and the Rayleigh–Jeans law for the Sun ($T_\odot = 5777$ K) on the same graph. At roughly what wavelength is the Rayleigh–Jeans value twice as large as the Planck function?

3.11 Show that Wien's expression for blackbody radiation (Eq. 3.21) follows directly from Planck's function at short wavelengths.

3.12 Derive Wien's displacement law, Eq. (3.15), by setting $dB_\lambda/d\lambda = 0$. *Hint:* You will encounter an equation that must be solved numerically, not algebraically.

3.13 **(a)** Use Eq. (3.24) to find an expression for the frequency $\nu_{\max}$ at which the Planck function B_ν attains its maximum value. (*Warning:* $\nu_{\max} \neq c/\lambda_{\max}$.)

(b) What is the value of $\nu_{\max}$ for the Sun?

(c) Find the wavelength of a light wave having frequency $\nu_{\max}$. In what region of the electromagnetic spectrum is this wavelength found?

3.14 **(a)** Integrate Eq. (3.27) over all wavelengths to obtain an expression for the total luminosity of a blackbody model star. *Hint:*

$$\int_0^\infty \frac{u^3\,du}{e^u - 1} = \frac{\pi^4}{15}.$$

(b) Compare your result with the Stefan–Boltzmann equation (3.17), and show that the Stefan–Boltzmann constant σ is given by

$$\sigma = \frac{2\pi^5 k^4}{15c^2h^3}.$$

(c) Calculate the value of σ from this expression, and compare with the value listed in Appendix A.

3.15 Use the data in Appendix G to answer the following questions.

(a) Calculate the absolute and apparent visual magnitudes, M_V and V, for the Sun.

(b) Determine the magnitudes M_B, B, M_U, and U for the Sun.

(c) Locate the Sun and Sirius on the color–color diagram in Fig. 3.11. Refer to Example 3.6.1 for the data on Sirius.

3.16 Use the filter bandwidths for the UBV system on page 75 and the effective temperature of 9600 K for Vega to determine through which filter Vega would appear brightest to a photometer [i.e., ignore the constant C in Eq. (3.31)]. Assume that $\mathcal{S}(\lambda) = 1$ inside the filter bandwidth and that $\mathcal{S}(\lambda) = 0$ outside the filter bandwidth.

3.17 Evaluate the constant $C_{\rm bol}$ in Eq. (3.32) by using $m_{\rm Sun} = -26.83$.

3.18 Use the values of the constants C_{U-B} and C_{B-V} found in Example 3.6.2 to estimate the color indices $U - B$ and $B - V$ for the Sun.

3.19 Shaula (λ Scorpii) is a bright ($V = 1.62$) blue-white subgiant star located at the tip of the scorpion's tail. Its surface temperature is about 22,000 K.

(a) Use the values of the constants C_{U-B} and C_{B-V} found in Example 3.6.2 to estimate the color indices $U - B$ and $B - V$ for Shaula. Compare your answers with the measured values of $U - B = -0.90$ and $B - V = -0.23$.

(b) The Hipparcos Space Astrometry Mission measured the parallax angle for Shaula to be $0.00464''$. Determine the absolute visual magnitude of the star.

(Shaula is a pulsating star, belonging to the class of Beta Cephei variables; see Section 14.2. As its magnitude varies between $V = 1.59$ and $V = 1.65$ with a period of 5 hours 8 minutes, its color indices also change slightly.)

CHAPTER 4 The Theory of Special Relativity

4.1 ■ THE FAILURE OF THE GALILEAN TRANSFORMATIONS

A *wave* is a disturbance that travels through a medium. Water waves are disturbances traveling through water, and sound waves are disturbances traveling through air. James Clerk Maxwell predicted that light consists of "modulations of the same medium which is the cause of electric and magnetic phenomena," but what was the medium through which light waves traveled? At the time, physicists believed that light waves moved through a medium called the **luminiferous ether**. This idea of an all-pervading ether had its roots in the science of early Greece. In addition to the four earthly elements of earth, air, fire, and water, the Greeks believed that the heavens were composed of a fifth perfect element: the ether. Maxwell echoed their ancient belief when he wrote:

> There can be no doubt that the interplanetary and interstellar spaces are not empty, but are occupied by a material substance or body, which is certainly the largest, and probably the most uniform body of which we have any knowledge.

This modern reincarnation of the ether had been proposed for the sole purpose of transporting light waves; an object moving through the ether would experience no mechanical resistance, so Earth's velocity through the ether could not be directly measured.

The Galilean Transformations

In fact, *no* mechanical experiment is capable of determining the absolute velocity of an observer. It is impossible to tell whether you are at rest or in uniform motion (not accelerating). This general principle was recognized very early. Galileo described a laboratory completely enclosed below the deck of a smoothly sailing ship and argued that no experiment done in this uniformly moving laboratory could measure the ship's velocity. To see why, consider two *inertial reference frames*, S and S'. As discussed in Section 2.2, an inertial reference frame may be thought of as a laboratory in which Newton's first law is valid: An object at rest will remain at rest, and an object in motion will remain in motion in a straight line at

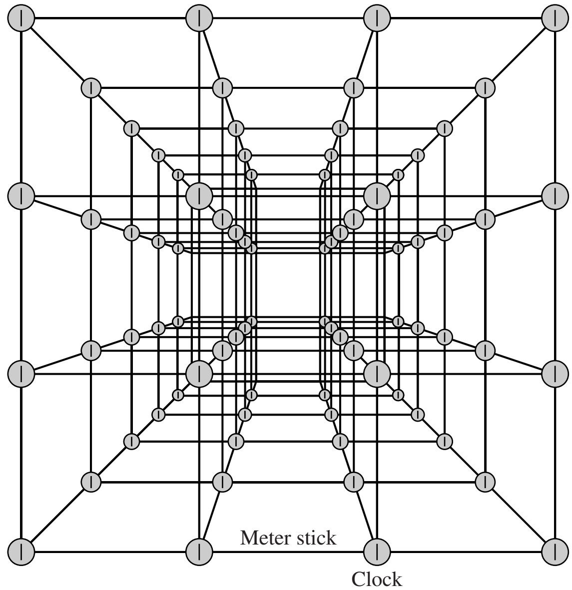

FIGURE 4.1 Inertial reference frame.

constant speed, unless acted upon by an external force. As shown in Fig. 4.1, the laboratory consists of (in principle) an infinite collection of meter sticks and synchronized clocks that can record the position and time of any event that occurs in the laboratory, *at the location of that event*; this removes the time delay involved in relaying information about an event to a distant recording device. With no loss of generality, the frame S' can be taken as moving in the positive x-direction (relative to the frame S) with constant velocity **u**, as shown in Fig. 4.2.[1] Furthermore, the clocks in the two frames can be started when the origins of the coordinate systems, O and O', coincide at time $t = t' = 0$.

Observers in the two frames S and S' measure the same moving object, recording its positions (x, y, z) and (x', y', z') at time t and t', respectively. An appeal to common sense and intuition leads to the conclusion that these measurements are related by the **Galilean transformation equations**:

$$x' = x - ut \tag{4.1}$$

$$y' = y \tag{4.2}$$

$$z' = z \tag{4.3}$$

$$t' = t. \tag{4.4}$$

[1]This does *not* imply that the frame S is at rest and that S' is moving. S' could be at rest while S moves in the negative x'-direction, or both frames could be moving. The point of the following argument is that there is *no way to tell*; only the *relative velocity* **u** is meaningful.

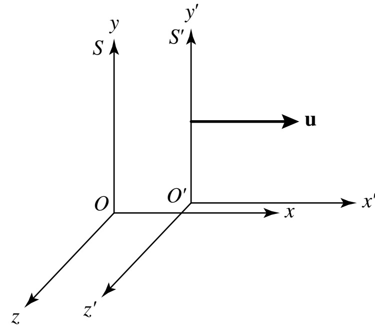

FIGURE 4.2 Inertial reference frames S and S'.

Taking time derivatives with respect to either t or t' (since they are always equal) shows how the components of the object's velocity $\mathbf{v}$ and $\mathbf{v}'$ measured in the two frames are related:

$$v'_x = v_x - u$$
$$v'_y = v_y$$
$$v'_z = v_z,$$

or, in vector form,

$$\mathbf{v}' = \mathbf{v} - \mathbf{u}. \tag{4.5}$$

Since $\mathbf{u}$ is constant, another time derivative shows that the *same* acceleration is obtained for the object as measured in both reference frames:

$$\mathbf{a}' = \mathbf{a}.$$

Thus $\mathbf{F} = m\mathbf{a} = m\mathbf{a}'$ for the object of mass m; Newton's laws are obeyed in both reference frames. Whether a laboratory is located in the hold of Galileo's ship or anywhere else in the universe, no mechanical experiment can be done to measure the laboratory's absolute velocity.

The Michelson–Morley Experiment

Maxwell's discovery that electromagnetic waves move through the ether with a speed of $c \simeq 3 \times 10^8$ m s^{-1} seemed to open the possibility of detecting Earth's *absolute* motion through the ether by measuring the speed of light from Earth's frame of reference and comparing it with Maxwell's theoretical value of c. In 1887 two Americans, the physicist Albert A. Michelson (1852–1931) and his chemist colleague Edward W. Morley (1838–1923), performed a classic experiment that attempted this measurement of Earth's absolute velocity. Although Earth orbits the Sun at approximately 30 km s^{-1}, the results of the Michelson–Morley experiment were consistent with a velocity of Earth through the ether of *zero*![2] Furthermore, as Earth spins on its axis and orbits the Sun, a laboratory's speed through the ether should be constantly changing. The constantly shifting "ether wind" should easily be detected. However, all of the many physicists who have since repeated the Michelson–Morley experiment with increasing precision have reported the same null result. Everyone measures *exactly* the same value for the speed of light, regardless of the velocity of the laboratory on Earth or the velocity of the source of the light.

On the other hand, Eq. (4.5) implies that two observers moving with a relative velocity $\mathbf{u}$ should obtain *different* values for the speed of light. The contradiction between the commonsense expectation of Eq. (4.5) and the experimentally determined constancy of the speed of light means that this equation, and the equations from which it was derived (the Galilean transformation equations, 4.1–4.4), cannot be correct. Although the Galilean transformations adequately describe the familiar low-speed world of everyday life where $v/c \ll 1$, they are in sharp disagreement with the results of experiments involving velocities near the speed of light. A crisis in the Newtonian paradigm was developing.

4.2 ■ THE LORENTZ TRANSFORMATIONS

The young Albert Einstein (1879–1955; see Fig. 4.3) enjoyed discussing a puzzle with his friends: What would you see if you looked in a mirror while moving at the speed of light? Would you see your image in the mirror, or not? This was the beginning of Einstein's search for a simple, consistent picture of the universe, a quest that would culminate in his theories

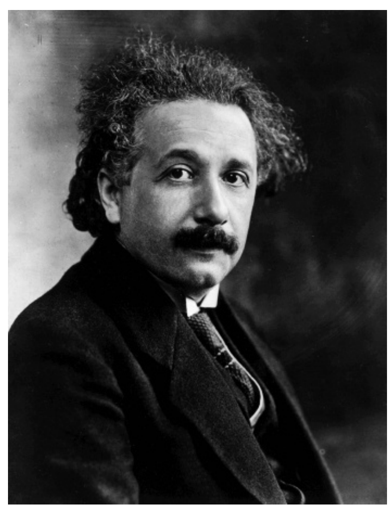

FIGURE 4.3 Albert Einstein (1879–1955). (Courtesy of Yerkes Observatory.)

[2]Strictly speaking, a laboratory on Earth is not in an inertial frame of reference, because Earth both spins on its axis and accelerates as it orbits the Sun. However, these noninertial effects are unimportant for the Michelson–Morley experiment.

of relativity. After much reflection, Einstein finally rejected the notion of an all-pervading ether.

Einstein's Postulates

In 1905 Einstein introduced his two postulates of special relativity[3] in a remarkable paper, "On the Electrodynamics of Moving Bodies."

> The phenomena of electrodynamics as well as of mechanics possess no properties corresponding to the idea of absolute rest. They suggest rather that … the same laws of electrodynamics and optics will be valid for all frames of reference for which the equations of mechanics hold good. We will raise this conjecture (the purport of which will hereafter be called the "Principle of Relativity") to the status of a postulate, and also introduce another postulate, which is only apparently irreconcilable to the former, namely, that light is always propagated in empty space with a definite speed c which is independent of the state of motion of the emitting body.

In other words, **Einstein's postulates** are

> **The Principle of Relativity** The laws of physics are the same in all inertial reference frames.
>
> **The Constancy of the Speed of Light** Light moves through a vacuum at a constant speed c that is independent of the motion of the light source.

The Derivation of the Lorentz Transformations

Einstein then went on to derive the equations that lie at the heart of his theory of special relativity, the **Lorentz transformations**.[4] For the two inertial reference frames shown in Fig. 4.2, the most general set of linear transformation equations between the space and time coordinates (x, y, z, t) and (x', y', z', t') of the *same event* measured from S and S' are

$$x' = a_{11}x + a_{12}y + a_{13}z + a_{14}t \tag{4.6}$$

$$y' = a_{21}x + a_{22}y + a_{23}z + a_{24}t \tag{4.7}$$

$$z' = a_{31}x + a_{32}y + a_{33}z + a_{34}t \tag{4.8}$$

$$t' = a_{41}x + a_{42}y + a_{43}z + a_{44}t. \tag{4.9}$$

If the transformation equations were not linear, then the length of a moving object or the time interval between two events would depend on the choice of origin for the frames S and S'. This is unacceptable, since the laws of physics cannot depend on the numerical coordinates of an arbitrarily chosen coordinate system.

[3]The theory of *special* relativity deals only with inertial reference frames, whereas the *general* theory includes accelerating frames.

[4]These equations were first derived by Hendrik A. Lorentz (1853–1928) of the Netherlands but were applied to a different situation involving a reference frame at absolute rest with respect to the ether.

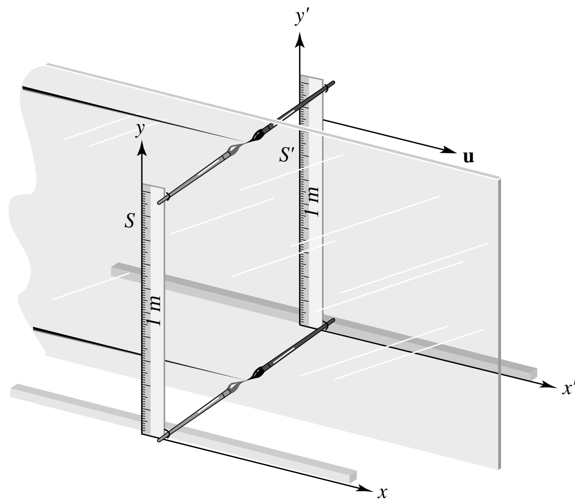

FIGURE 4.4 Paint brush demonstration that $y' = y$.

The coefficients a_{ij} can be determined by using Einstein's two postulates and some simple symmetry arguments. Einstein's first postulate, the Principle of Relativity, implies that lengths *perpendicular* to **u**, the velocity of frame S relative to S', are unchanged. To see this, imagine that each frame has a meter stick oriented along the y- and y'-axes, with one end of each meter stick located at the origin of its respective frame; see Fig. 4.4. Paint brushes are mounted perpendicular at both ends of each meter stick, and the frames are separated by a sheet of glass that extends to infinity in the x–y plane. Each brush paints a line on the glass sheet as the two frames pass each other. Let's say that frame S uses blue paint, and frame S' uses red paint. If an observer in frame S measures the meter stick in frame S' to be shorter than his own meter stick, he will see the red lines painted *inside* his blue lines on the glass. But by the Principle of Relativity, an observer in frame S' would measure the meter stick in frame S as being shorter than her own meter stick and would see the blue lines painted *inside* her red lines. Both color lines cannot lie inside the other; the only conclusion is that blue and red lines must overlap. The lengths of the meter sticks, perpendicular to **u**, are unchanged. Thus $y' = y$ and $z' = z$, so that $a_{22} = a_{33} = 1$, whereas $a_{21},\ a_{23},\ a_{24},\ a_{31},\ a_{32}$, and a_{34} are all zero.

Another simplification comes from requiring that Eq. (4.9) give the same result if y is replaced by $-y$ or z is replaced by $-z$. This must be true because rotational symmetry about the axis parallel to the relative velocity **u** implies that a time measurement cannot depend on the side of the x-axis on which an event occurs. Thus $a_{42} = a_{43} = 0$.

Finally, consider the motion of the origin O' of frame S'. Since the frames' clocks are assumed to be synchronized at $t = t' = 0$ when the origins O and O' coincide, the x-coordinate of O' is given by $x = ut$ in frame S and by $x' = 0$ in frame S'. Thus Eq. (4.6) becomes

$$0 = a_{11}ut + a_{12}y + a_{13}z + a_{14}t,$$

which implies that $a_{12} = a_{13} = 0$ and $a_{11}u = -a_{14}$. Collecting the results found thus far reveals that Eqs. (4.6–4.9) have been reduced to

$$x' = a_{11}(x - ut) \tag{4.10}$$

$$y' = y \tag{4.11}$$

$$z' = z \tag{4.12}$$

$$t' = a_{41}x + a_{44}t. \tag{4.13}$$

At this point, these equations would be consistent with the commonsense Galilean transformation equations (4.1–4.4) if $a_{11} = a_{44} = 1$ and $a_{41} = 0$. However, only one of Einstein's postulates has been employed in the derivation thus far: the Principle of Relativity championed by Galileo himself.

Now the argument introduces the second of Einstein's postulates: Everyone measures exactly the same value for the speed of light. Suppose that when the origins O and O' coincide at time $t = t' = 0$, a flashbulb is set off at the common origins. At a later time t, an observer in frame S will measure a spherical wavefront of light with radius ct, moving away from the origin O with speed c and satisfying

$$x^2 + y^2 + z^2 = (ct)^2. \tag{4.14}$$

Similarly, at a time t', an observer in frame S' will measure a spherical wavefront of light with radius ct', moving away from the origin O' with speed c and satisfying

$$x'^2 + y'^2 + z'^2 = \left(ct'\right)^2. \tag{4.15}$$

Inserting Eqs. (4.10–4.13) into Eq. (4.15) and comparing the result with Eq. (4.14) reveal that $a_{11} = a_{44} = 1\big/\sqrt{1 - u^2/c^2}$ and $a_{41} = -ua_{11}/c^2$. Thus the Lorentz transformation equations linking the space and time coordinates (x, y, z, t) and (x', y', z', t') of the *same event* measured from S and S' are

$$\boxed{\begin{aligned} x' &= \frac{x - ut}{\sqrt{1 - u^2/c^2}} && (4.16)\\ y' &= y && (4.17)\\ z' &= z && (4.18)\\ t' &= \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}}. && (4.19) \end{aligned}}$$

Whenever the Lorentz transformations are used, you should be certain that the situation is consistent with the geometry of Fig. 4.2, where the inertial reference frame S' is moving in

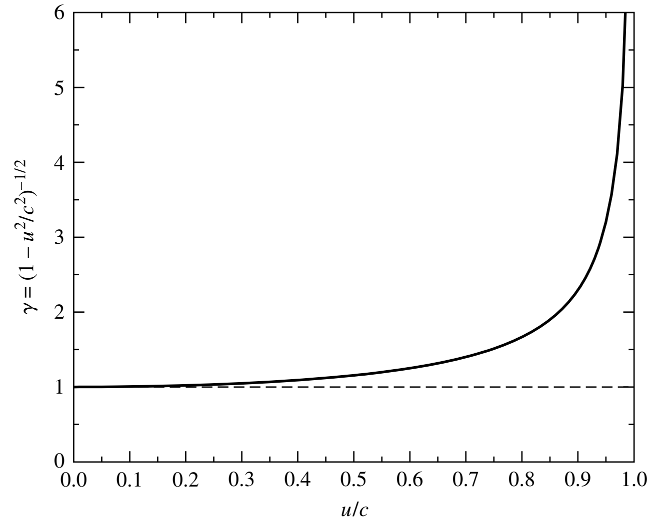

FIGURE 4.5 The Lorentz factor γ.

the positive x-direction with velocity $\mathbf{u}$ relative to the frame S. The ubiquitous factor of

$$\boxed{\gamma \equiv \frac{1}{\sqrt{1 - u^2/c^2}},} \tag{4.20}$$

called the **Lorentz factor**, may be used to estimate the importance of relativistic effects. Roughly speaking, relativity differs from Newtonian mechanics by 1% ($\gamma = 1.01$) when $u/c \simeq 1/7$ and by 10% when $u/c \simeq 5/12$; see Fig. 4.5. In the low-speed Newtonian world, the Lorentz transformations reduce to the Galilean transformation equations (4.1–4.4). A similar requirement holds for all relativistic formulas; they must agree with the Newtonian equations in the low-speed limit of $u/c \to 0$.

The inverse Lorentz transformations can be derived algebraically, or they can be obtained more easily by switching primed and unprimed quantities and by replacing u with $-u$. (Be sure you understand the physical basis for these substitutions.) Either way, the inverse transformations are found to be

$$x = \frac{x' + ut'}{\sqrt{1 - u^2/c^2}} \tag{4.21}$$

$$y = y' \tag{4.22}$$

$$z = z' \tag{4.23}$$

$$t = \frac{t' + ux'/c^2}{\sqrt{1 - u^2/c^2}}. \tag{4.24}$$

Four-Dimensional Spacetime

The Lorentz transformation equations form the core of the theory of special relativity, and they have many surprising and unusual implications. The most obvious surprise is the intertwining roles of spatial and temporal coordinates in the transformations. In the words of Einstein's professor, Hermann Minkowski (1864–1909), "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows, and only a kind of union between the two will preserve an independent reality." The drama of the physical world unfolds on the stage of a four-dimensional **spacetime**, where events are identified by their spacetime coordinates (x, y, z, t).

4.3 ■ TIME AND SPACE IN SPECIAL RELATIVITY

Suppose an observer in frame S measures two flashbulbs going off at the *same time* t but at *different* x-coordinates x_1 and x_2. Then an observer in frame S' would measure the time interval $t_1' - t_2'$ between the flashbulbs going off to be (see Eq. 4.19)

$$t_1' - t_2' = \frac{(x_2 - x_1)\, u/c^2}{\sqrt{1 - u^2/c^2}}. \tag{4.25}$$

According to the observer in frame S', if $x_1 \neq x_2$, then the flashbulbs do not go off at the same time! Events that occur simultaneously in one inertial reference frame do not occur simultaneously in all other inertial reference frames. There is no such thing as two events that occur at different locations happening *absolutely* at the same time. Equation (4.25) shows that if $x_1 < x_2$, then $t_1' - t_2' > 0$ for positive u; flashbulb 1 is measured to go off *after* flashbulb 2. An observer moving at the same speed in the opposite direction (u changed to $-u$) will come to the opposite conclusion: Flashbulb 2 goes off *after* flashbulb 1. The situation is symmetric; an observer in frame S' will conclude that the flashbulb he or she passes first goes off *after* the other flashbulb. It is tempting to ask, "Which observer is *really* correct?" However, this question is meaningless and is equivalent to asking, "Which observer is *really* moving?" Neither question has an answer because "really" has no meaning in this situation. There is no absolute simultaneity, just as there is no absolute motion. Each observer's measurement is correct, as made from his or her own frame of reference.

The implications of this **downfall of universal simultaneity** are far-reaching. The absence of a universal simultaneity means that clocks in relative motion will not stay synchronized. Newton's idea of an absolute universal time that "of itself and from its own nature flows equably without regard to anything external" has been overthrown. Different observers in relative motion will measure *different* time intervals between the *same* two events!

Proper Time and Time Dilation

Imagine that a strobe light located at rest relative to frame S' produces a flash of light every $\Delta t'$ seconds; see Fig. 4.6. If one flash is emitted at time t_1', then the next flash will be emitted at time $t_2' = t_1' + \Delta t'$, as measured by a clock in frame S'. Using Eq. (4.24) with $x_1' = x_2'$,

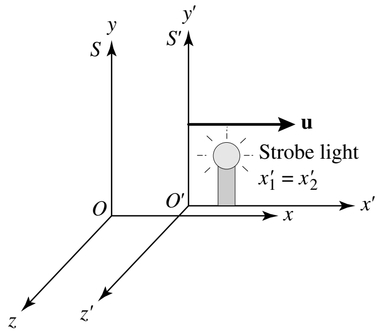

FIGURE 4.6 A strobe light at rest ($x' =$ constant) in frame S'.

the time interval $\Delta t \equiv t_2 - t_1$ between the *same* two flashes measured by a clock in frame S is

$$t_2 - t_1 = \frac{(t_2' - t_1') + (x_2' - x_1')\,u/c^2}{\sqrt{1 - u^2/c^2}}$$

or

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - u^2/c^2}}. \tag{4.26}$$

Because the clock in frame S' is *at rest* relative to the strobe light, $\Delta t'$ will be called Δt_{rest}. Frame S' is called the clock's **rest frame**. Similarly, because the clock in frame S is *moving* relative to the strobe light, Δt will be called Δt_{moving}. Thus Eq. (4.26) becomes

$$\boxed{\Delta t_{\text{moving}} = \frac{\Delta t_{\text{rest}}}{\sqrt{1 - u^2/c^2}}.} \tag{4.27}$$

This equation shows the effect of **time dilation** on a moving clock. It says that the time interval between two events is measured differently by different observers in relative motion. The *shortest time interval* is measured by a clock *at rest* relative to the two events. This clock measures the **proper time** between the two events. Any other clock moving relative to the two events will measure a longer time interval between them.

The effect of time dilation is often described by the phrase "moving clocks run slower" without explicitly identifying the two events involved. This easily leads to confusion, since the moving and rest subscripts in Eq. (4.27) mean "moving" or "at rest" *relative to the two events*. To gain insight into this phrase, imagine that you are holding clock C while it ticks once each second and, at the same time, are measuring the ticks of an identical clock C' moving relative to you. The two events to be measured are consecutive ticks of clock C'. Since clock C' is at rest relative to itself, it measures a time $\Delta t_{\text{rest}} = 1$ s between its own ticks. However, using your clock C, you measure a time

$$\Delta t_{\text{moving}} = \frac{\Delta t_{\text{rest}}}{\sqrt{1 - u^2/c^2}} = \frac{1\ \text{s}}{\sqrt{1 - u^2/c^2}} > 1\ \text{s}$$

between the ticks of clock C'. Because you measure clock C' to be ticking slower than once per second, you conclude that clock C', which is moving relative to you, is running more slowly than your clock C. Very accurate atomic clocks have been flown around the world on jet airliners and have confirmed that moving clocks do indeed run slower, in agreement with relativity.[5]

Proper Length and Length Contraction

Both time dilation and the downfall of simultaneity contradict Newton's belief in absolute time. Instead, the time measured between two events differs for different observers in relative motion. Newton also believed that "absolute space, in its own nature, without relation to anything external, remains always similar and immovable." However, the Lorentz transformation equations require that different observers in relative motion will measure space differently as well.

Imagine that a rod lies along the x'-axis of frame S', at rest relative to that frame; S' is the rod's rest frame (see Fig. 4.7). Let the left end of the rod have coordinate x'_1, and let the right end of the rod have coordinate x'_2. Then the length of the rod as measured in frame S' is $L' = x'_2 - x'_1$. What is the length of the rod measured from S? Because the rod is moving relative to S, care must be taken to measure the x-coordinates x_1 and x_2 of the ends of the rod *at the same time*. Then Eq. (4.16), with $t_1 = t_2$, shows that the length $L = x_2 - x_1$ measured in S may be found from

$$x'_2 - x'_1 = \frac{(x_2 - x_1) - u(t_2 - t_1)}{\sqrt{1 - u^2/c^2}}$$

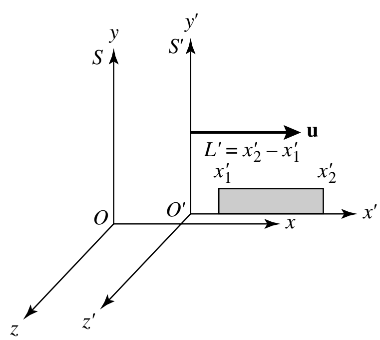

FIGURE 4.7 A rod at rest in frame S'.

[5]See Hafele and Keating (1972a, 1972b) for the details of this test of time dilation.

or

$$L' = \frac{L}{\sqrt{1 - u^2/c^2}}. \tag{4.28}$$

Because the rod is *at rest* relative to S', L' will be called L_{rest}. Similarly, because the rod is *moving* relative to S, L will be called L_{moving}. Thus Eq. (4.28) becomes

$$\boxed{L_{\text{moving}} = L_{\text{rest}}\sqrt{1 - u^2/c^2}.} \tag{4.29}$$

This equation shows the effect of **length contraction** on a moving rod. It says that length or distance is measured differently by two observers in relative motion. If a rod is moving relative to an observer, that observer will measure a shorter rod than will an observer at rest relative to it. The *longest length*, called the rod's **proper length**, is measured in the rod's rest frame. Only lengths or distances *parallel* to the direction of the relative motion are affected by length contraction; distances perpendicular to the direction of the relative motion are unchanged (cf. Eqs. 4.17–4.18).

Time Dilation and Length Contraction Are Complementary

Time dilation and length contraction are not independent effects of Einstien's new way of looking at the universe. Rather, they are complementary; the magnitude of either effect depends on the motion of the event being observed relative to the observer.

Example 4.3.1. Cosmic rays from space collide with the nuclei of atoms in Earth's upper atmosphere, producing elementary particles called *muons*. Muons are unstable and decay after an average lifetime $\tau = 2.20\ \mu$s, as measured in a laboratory where the muons are at rest. That is, the number of muons in a given sample should decrease with time according to $N(t) = N_0\, e^{-t/\tau}$, where N_0 is the number of muons originally in the sample at time $t = 0$. At the top of Mt. Washington in New Hampshire, a detector counted 563 muons hr^{-1} moving downward at a speed $u = 0.9952c$. At sea level, 1907 m below the first detector, another detector counted 408 muons hr^{-1}.[6]

The muons take $(1907\text{ m})/(0.9952c) = 6.39\ \mu$s to travel from the top of Mt. Washington to sea level. Thus it might be expected that the number of muons detected per hour at sea level would have been

$$N = N_0\, e^{-t/\tau} = (563\text{ muons hr}^{-1})\, e^{-(6.39\ \mu\text{s})/(2.20\ \mu\text{s})} = 31\text{ muons hr}^{-1}.$$

This is much less than the 408 muons hr^{-1} actually measured at sea level! How did the muons live long enough to reach the lower detector? The problem with the preceding calculation is that the lifetime of 2.20 μs is measured in the muon's rest frame as Δt_{rest}, but the experimenter's clocks on Mt. Washington and below are moving relative to the muons.

continued

[6]Details of this experiment can be found in Frisch and Smith (1963).

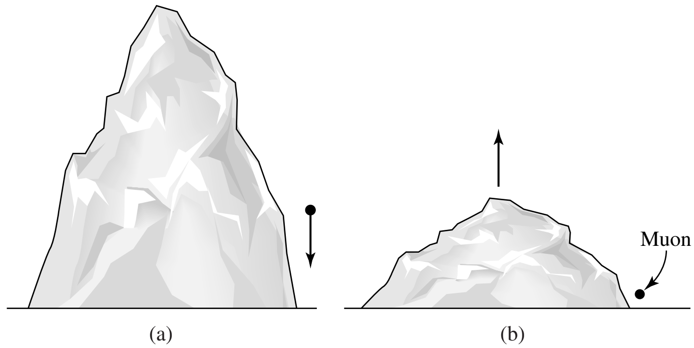

FIGURE 4.8 Muons moving downward past Mt. Washington. (a) Mountain frame. (b) Muon frame.

They measure the muon's lifetime to be

$$\Delta t_{\text{moving}} = \frac{\Delta t_{\text{rest}}}{\sqrt{1 - u^2/c^2}} = \frac{2.20\ \mu\text{s}}{\sqrt{1 - (0.9952)^2}} = 22.5\ \mu\text{s},$$

more than *ten* times a muon's lifetime when measured in its own rest frame. The moving muons' clocks run slower, so more of them survive long enough to reach sea level. Repeating the preceding calculation using the muon lifetime as measured by the experimenters gives

$$N = N_0\, e^{-t/\tau} = (563 \text{ muons hr}^{-1})\, e^{-(6.39\ \mu\text{s})/(22.5\ \mu\text{s})} = 424 \text{ muons hr}^{-1}.$$

When the effects of time dilation are included, the theoretical prediction is in excellent agreement with the experimental result.

From a muon's rest frame, its lifetime is only 2.20 μs. How would an observer riding along with the muons, as shown in Fig. 4.8, explain their ability to reach sea level? The observer would measure a severely length-contracted Mt. Washington (in the direction of the relative motion only). The distance traveled by the muons would not be $L_{\text{rest}} = 1907$ m but, rather, would be

$$L_{\text{moving}} = L_{\text{rest}}\sqrt{1 - u^2/c^2} = (1907 \text{ m})\sqrt{1 - (0.9952)^2} = 186.6 \text{ m}.$$

Thus it would take $(186.6 \text{ m})/(0.9952c) = 0.625\ \mu$s for the muons to travel the length-contracted distance to the detector at sea level, as measured by an observer in the muons' rest frame. That observer would then calculate the number of muons reaching the lower detector to be

$$N = N_0\, e^{-t/\tau} = (563 \text{ muons hr}^{-1})\, e^{-(0.625\ \mu\text{s})/(2.20\ \mu\text{s})} = 424 \text{ muons hr}^{-1},$$

in agreement with the previous result. This shows that an effect due to time dilation as measured in one frame may instead be attributed to length contraction as measured in another frame.

The effects of time dilation and length contraction are both symmetric between two observers in relative motion. Imagine two identical spaceships that move in opposite directions, passing each other at some relativistic speed. Observers aboard each spaceship will measure the other ship's length as being shorter than their own, and the other ship's clocks as running slower. *Both observers are right*, having made correct measurements from their respective frames of reference.

You should not think of these effects as being due to some sort of "optical illusion" caused by light taking different amounts of time to reach an observer from different parts of a moving object. The language used in the preceding discussions has involved the *measurement* of an event's spacetime coordinates (x, y, z, t) using meter sticks and clocks located *at that event*, so there is no time delay. Of course, no actual laboratory has an infinite collection of meter sticks and clocks, and the time delays caused by finite light-travel times must be taken into consideration. This will be important in determining the relativistic Doppler shift formula, which follows.

The Relativistic Doppler Shift

In 1842 the Austrian physicist Christian Doppler showed that as a source of sound moves through a medium (such as air), the wavelength is compressed in the forward direction and expanded in the backward direction. This change in wavelength of any type of wave caused by the motion of the source or the observer is called a **Doppler shift**. Doppler deduced that the difference between the wavelength λ_{obs} observed for a moving source of sound and the wavelength λ_{rest} measured in the laboratory for a reference source at rest is related to the radial velocity v_r (the component of the velocity directly toward or away from the observer; see Fig. 1.15) of the source through the medium by

$$\frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = \frac{\Delta\lambda}{\lambda_{\text{rest}}} = \frac{v_r}{v_s}, \tag{4.30}$$

where v_s is the speed of sound in the medium. However, this expression cannot be precisely correct for light. Experimental results such as those of Michelson and Morley led Einstein to abandon the ether concept, and they demonstrated that no medium is involved in the propagation of light waves. The Doppler shift for light is a qualitatively different phenomenon from its counterpart for sound waves.

Consider a distant source of light that emits a light signal at time $t_{\text{rest},1}$ and another signal at time $t_{\text{rest},2} = t_{\text{rest},1} + \Delta t_{\text{rest}}$ as measured by a clock *at rest* relative to the source. If this light source is moving relative to an observer with velocity $\mathbf{u}$, as shown in Fig. 4.9, then the time between receiving the light signals at the observer's location will depend on both the effect of time dilation and the different distances traveled by the signals from the source to the observer. (The light source is assumed to be sufficiently far away that the signals travel along parallel paths to the observer.) Using Eq. (4.27), we find that the time between the emission of the light signals as measured in the observer's frame is $\Delta t_{\text{rest}} \big/ \sqrt{1 - u^2/c^2}$. In this time, the observer determines that the distance to the light source has changed by an amount $u \Delta t_{\text{rest}} \cos\theta \big/ \sqrt{1 - u^2/c^2}$. Thus the time interval Δt_{obs} between the arrival of the

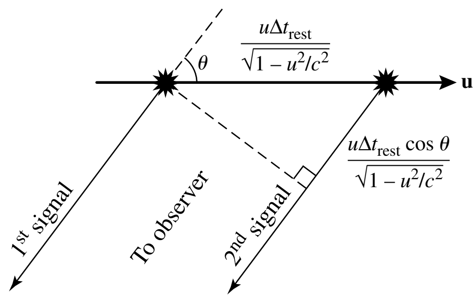

FIGURE 4.9 Relativistic Doppler shift.

two light signals at the observer's location is

$$\Delta t_{\text{obs}} = \frac{\Delta t_{\text{rest}}}{\sqrt{1-u^2/c^2}}\,[1+(u/c)\cos\theta]. \tag{4.31}$$

If Δt_{rest} is taken to be the time between the emission of the light wave crests, and if Δt_{obs} is the time between their arrival, then the frequencies of the light wave are $\nu_{\text{rest}} = 1/\Delta t_{\text{rest}}$ and $\nu_{\text{obs}} = 1/\Delta t_{\text{obs}}$. The equation describing the **relativistic Doppler shift** is thus

$$\boxed{\nu_{\text{obs}} = \frac{\nu_{\text{rest}}\sqrt{1-u^2/c^2}}{1+(u/c)\cos\theta} = \frac{\nu_{\text{rest}}\sqrt{1-u^2/c^2}}{1+v_r/c},} \tag{4.32}$$

where $v_r = u\cos\theta$ is the *radial velocity* of the light source. If the light source is moving directly away from the observer ($\theta = 0°$, $v_r = u$) or toward the observer ($\theta = 180°$, $v_r = -u$), then the relativistic Doppler shift reduces to

$$\nu_{\text{obs}} = \nu_{\text{rest}}\sqrt{\frac{1-v_r/c}{1+v_r/c}} \qquad \text{(radial motion)}. \tag{4.33}$$

There is also a **transverse Doppler shift** for motion perpendicular to the observer's line of sight ($\theta = 90°$, $v_r = 0$). This transverse shift is entirely due to the effect of time dilation. Note that, unlike formulas describing the Doppler shift for sound, Eqs. (4.32) and (4.33) do not distinguish between the velocity of the source and the velocity of the observer. Only the relative velocity is important.

When astronomers observe a star or galaxy moving away from or toward Earth, the wavelength of the light they receive is shifted toward longer or shorter wavelengths, respectively. If the source of light is moving *away* from the observer ($v_r > 0$), then $\lambda_{\text{obs}} > \lambda_{\text{rest}}$. This shift to a longer wavelength is called a **redshift**. Similarly, if the source is moving *toward* the observer ($v_r < 0$), then there is a shift to a shorter wavelength, a **blueshift**.[7] Because

[7]Doppler himself maintained that all stars would be white if they were at rest and that the different colors of the stars were due to their Doppler shifts. However, the stars move much too slowly for their Doppler shifts to significantly change their colors.

most of the objects in the universe outside of our own Milky Way Galaxy are moving away from us, redshifts are commonly measured by astronomers. A **redshift parameter** z is used to describe the change in wavelength; it is defined as

$$\boxed{z \equiv \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = \frac{\Delta\lambda}{\lambda_{\text{rest}}}.} \tag{4.34}$$

The observed wavelength λ_{obs} is obtained from Eq. (4.33) and $c = \lambda\nu$,

$$\lambda_{\text{obs}} = \lambda_{\text{rest}}\sqrt{\frac{1 + v_r/c}{1 - v_r/c}} \qquad \text{(radial motion)}, \tag{4.35}$$

and the redshift parameter becomes

$$\boxed{z = \sqrt{\frac{1 + v_r/c}{1 - v_r/c}} - 1 \qquad \text{(radial motion).}} \tag{4.36}$$

In general, Eq. (4.34), together with $\lambda = c/\nu$, shows that

$$\boxed{z + 1 = \frac{\Delta t_{\text{obs}}}{\Delta t_{\text{rest}}}.} \tag{4.37}$$

This expression indicates that if the luminosity of an astrophysical source with redshift parameter $z > 0$ (receding) is observed to vary during a time Δt_{obs}, then the change in luminosity occurred over a *shorter* time $\Delta t_{\text{rest}} = \Delta t_{\text{obs}}/(z + 1)$ in the rest frame of the source.

Example 4.3.2. In its rest frame, the quasar SDSS 1030+0524 produces a hydrogen emission line of wavelength $\lambda_{\text{rest}} = 121.6$ nm. On Earth, this emission line is observed to have a wavelength of $\lambda_{\text{obs}} = 885.2$ nm. The redshift parameter for this quasar is thus

$$z = \frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = 6.28.$$

Using Eq. (4.36), we may calculate the speed of recession of the quasar:

$$\begin{aligned} z &= \sqrt{\frac{1 + v_r/c}{1 - v_r/c}} - 1 \\ \frac{v_r}{c} &= \frac{(z+1)^2 - 1}{(z+1)^2 + 1} \qquad (4.38) \\ &= 0.963. \end{aligned}$$

continued

Quasar SDSS 1030+0524 appears to be moving away from us at more than 96% of the speed of light! However, objects that are enormously distant from us, such as quasars, have large apparent recessional speeds due to the overall expansion of the universe. In these cases the increase in the observed wavelength is actually due to *the expansion of space itself* (which stretches the wavelength of light) rather than being due to the motion of the object through space! This *cosmological redshift* is a consequence of the Big Bang.

This quasar was discovered as a product of the massive Sloan Digital Sky Survey; see Becker, et al. (2001) for further information about this object.

Suppose the speed u of the light source is small compared to that of light ($u/c \ll 1$). Using the expansion (to first order)

$$(1 + v_r/c)^{\pm 1/2} \simeq 1 \pm \frac{v_r}{2c},$$

together with Eqs. (4.34) and (4.35) for radial motion, then shows that for low speeds,

$$z = \frac{\Delta\lambda}{\lambda_{\text{rest}}} \simeq \frac{v_r}{c}, \tag{4.39}$$

where $v_r > 0$ for a receding source ($\Delta\lambda > 0$) and $v_r < 0$ for an approaching source ($\Delta\lambda < 0$). Although this equation is similar to Eq. (4.30), you should bear in mind that Eq. (4.39) is an *approximation*, valid only for low speeds. Misapplying this equation to the relativistic quasar SDSS 1030+0524 discussed in Example 4.3.2 would lead to the erroneous conclusion that the quasar is moving away from us at 6.28 times the speed of light!

The Relativistic Velocity Transformation

Because space and time intervals are measured differently by different observers in relative motion, velocities must be transformed as well. The equations describing the relativistic transformation of velocities may be easily found from the Lorentz transformation equations (4.16–4.19) by writing them as differentials. Then dividing the dx', dy', and dz' equations by the dt' equation gives the **relativistic velocity transformations**:

$$\boxed{\begin{aligned} v_x' &= \frac{v_x - u}{1 - u v_x/c^2} &(4.40)\\ v_y' &= \frac{v_y\sqrt{1 - u^2/c^2}}{1 - u v_x/c^2} &(4.41)\\ v_z' &= \frac{v_z\sqrt{1 - u^2/c^2}}{1 - u v_x/c^2}. &(4.42)\end{aligned}}$$

As with the inverse Lorentz transformations, the inverse velocity transformations may be obtained by switching primed and unprimed quantities and by replacing u with $-u$. It is left as an exercise to show that these equations do satisfy the second of Einstein's postulates: Light travels through a vacuum at a constant speed that is independent of the motion of the light source. From Eqs. (4.40–4.42), if $\mathbf{v}$ has a magnitude of c, so does $\mathbf{v}'$ (see Problem 4.12).

Example 4.3.3. As measured in the reference frame S', a light source is at rest and radiates light equally in all directions. In particular, half of the light is emitted into the forward (positive x') hemisphere. Is this situation any different when viewed from frame S, which measures the light source traveling in the positive x-direction with a relativistic speed u?

Consider a light ray whose velocity components measured in S' are $v'_x = 0$, $v'_y = c$, and $v'_z = 0$. This ray travels along the boundary between the forward and backward hemispheres of light as measured in S'. However, as measured in frame S, this light ray has the velocity components given by the inverse transformations of Eqs. (4.40–4.42):

$$v_x = \frac{v'_x + u}{1 + uv'_x/c^2} = u$$

$$v_y = \frac{v'_y\sqrt{1 - u^2/c^2}}{1 + uv'_x/c^2} = c\sqrt{1 - u^2/c^2}$$

$$v_z = \frac{v'_z\sqrt{1 - u^2/c^2}}{1 + uv'_x/c^2} = 0.$$

As measured in frame S, the light ray is not traveling perpendicular to the x-axis; see Fig. 4.10.

In fact, for u/c close to 1, the angle θ measured between the light ray and the x-axis may be found from $\sin\theta = v_y/v$, where

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2} = c$$

is the speed of the light ray measured in frame S. Thus

$$\boxed{\sin\theta = \frac{v_y}{v} = \sqrt{1 - u^2/c^2} = \gamma^{-1},} \tag{4.43}$$

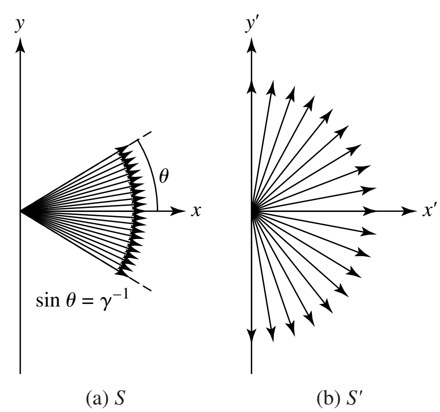

FIGURE 4.10 Relativistic headlight effect. (a) Frame S. (b) Frame S'.

continued

where γ is the Lorentz factor defined by Eq. (4.20). For relativistic speeds $u \approx c$, implying that γ is very large, so $\sin\theta$ (and hence θ) becomes very small. All of the light emitted into the forward hemisphere, as measured in S', is concentrated into a narrow cone in the direction of the light source's motion when measured in frame S. Called the **headlight effect**, this result plays an important role in many areas of astrophysics. For example, as relativistic electrons spiral around magnetic field lines, they emit light in the form of **synchrotron radiation**. The radiation is concentrated in the direction of the electron's motion and is strongly plane-polarized. Synchrotron radiation is an important electromagnetic radiation process in the Sun, Jupiter's magnetosphere, pulsars, and active galaxies.

4.4 ■ RELATIVISTIC MOMENTUM AND ENERGY

Up to this point, only relativistic kinematics has been considered. Einstein's theory of special relativity also requires new definitions for the concepts of momentum and energy. The ideas of conservation of linear momentum and energy are two of the cornerstones of physics. According to the Principle of Relativity, if momentum is conserved in one inertial frame of reference, then it must be conserved in all inertial frames. At the end of this section, it is shown that this requirement leads to a definition of the **relativistic momentum vector p**:

$$\boxed{\mathbf{p} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}} = \gamma m\mathbf{v},} \tag{4.44}$$

where γ is the Lorentz factor defined by Eq. (4.20). *Warning:* Some authors prefer to separate the "m" and the "$\mathbf{v}$" in this formula by defining a "relativistic mass," $m\big/\sqrt{1 - v^2/c^2}$. There is no compelling reason for this separation, and it can be misleading. In this text, the mass m of a particle is taken to be the same value in *all* inertial reference frames; it is **invariant** under a Lorentz transformation, and so there is no reason to qualify the term as a "rest mass." Thus the mass of a moving particle *does not* increase with increasing speed, although its *momentum* approaches infinity as $v \to c$. Also note that the "v" in the denominator is the magnitude of the particle's velocity relative to the observer, *not* the relative velocity u between two arbitrary frames of reference.

The Derivation of $E = mc^2$

Using Eq. (4.44) and the relation between kinetic energy and work from Section 2.2, we can derive an expression for the relativistic kinetic energy. The starting point is Newton's second law, $\mathbf{F} = d\mathbf{p}/dt$, applied to a particle of mass m that is initially at rest.[8] Consider a force of magnitude F that acts on the particle in the x-direction. The particle's final kinetic energy K equals the total work done by the force on the particle as it travels from its initial

[8] It is left as an exercise to show that $\mathbf{F} = m\mathbf{a}$ is *not correct*, since at relativistic speeds the force and the acceleration need *not* be in the same direction!

position x_i to its final position x_f:

$$K = \int_{x_i}^{x_f} F\,dx = \int_{x_i}^{x_f} \frac{dp}{dt}\,dx = \int_{p_i}^{p_f} \frac{dx}{dt}\,dp = \int_{p_i}^{p_f} v\,dp,$$

where p_i and p_f are the initial and final momenta of the particle, respectively. Integrating the last expression by parts and using the initial condition $p_i = 0$ give

$$\begin{aligned} K &= p_f v_f - \int_0^{v_f} p\,dv \\ &= \frac{mv_f^2}{\sqrt{1 - v_f^2/c^2}} - \int_0^{v_f} \frac{mv}{\sqrt{1 - v^2/c^2}}\,dv \\ &= \frac{mv_f^2}{\sqrt{1 - v_f^2/c^2}} + mc^2\left(\sqrt{1 - v_f^2/c^2} - 1\right). \end{aligned}$$

If we drop the f subscript, the expression for the **relativistic kinetic energy** becomes

$$\boxed{K = mc^2\left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right) = mc^2(\gamma - 1).} \tag{4.45}$$

Although it is not apparent that this formula for the kinetic energy reduces to either of the familiar forms $K = \frac{1}{2}mv^2$ or $K = p^2/2m$ in the low-speed Newtonian limit, both forms must be true if Eq. (4.45) is to be correct. The proofs will be left as exercises.

The right-hand side of this expression for the kinetic energy consists of the difference between two energy terms. The first is identified as the **total relativistic energy** E,

$$\boxed{E = \frac{mc^2}{\sqrt{1 - v^2/c^2}} = \gamma mc^2.} \tag{4.46}$$

The second term is an energy that does not depend on the speed of the particle; the particle has this energy even when it is at rest. The term mc^2 is called the **rest energy** of the particle:

$$E_{\text{rest}} = mc^2. \tag{4.47}$$

The particle's kinetic energy is its total energy minus its rest energy. When the energy of a particle is given as (for example) 40 MeV, the implicit meaning is that the particle's *kinetic energy* is 40 MeV; the rest energy is not included. Finally, there is a very useful expression relating a particle's total energy E, the magnitude of its momentum p, and its rest energy mc^2. It states that

$$\boxed{E^2 = p^2c^2 + m^2c^4.} \tag{4.48}$$

As we will discuss in Section 5.2, this equation is valid even for particles that have no mass, such as photons.

For a *system* of n particles, the total energy E_{sys} of the system is the sum of the total energies E_i of the individual particles: $E_{\mathrm{sys}} = \sum_{i=1}^{n} E_i$. Similarly, the vector momentum $\mathbf{p}_{\mathrm{sys}}$ of the system is the sum of the momenta $\mathbf{p}_i$ of the individual particles: $\mathbf{p}_{\mathrm{sys}} = \sum_{i=1}^{n} \mathbf{p}_i$. If the momentum of the system of particles is conserved, then the total energy is also conserved, *even for inelastic collisions* in which the kinetic energy of the system, $K_{\mathrm{sys}} = \sum_{i=1}^{n} K_i$, is reduced. The kinetic energy lost in the inelastic collisions goes into increasing the rest energy, and hence the mass, of the particles. This increase in rest energy allows the total energy of the system to be conserved. Mass and energy are two sides of the same coin; one can be transformed into the other.

Example 4.4.1. In a one-dimensional completely inelastic collision, two identical particles of mass m and speed v approach each other, collide head-on, and merge to form a single particle of mass M. The initial energy of the system of particles is

$$E_{\mathrm{sys},i} = \frac{2mc^2}{\sqrt{1 - v^2/c^2}}.$$

Since the initial momenta of the particles are equal in magnitude and opposite in direction, the momentum of the system $\mathbf{p}_{\mathrm{sys}} = 0$ before and after the collision. Thus after the collision, the particle is at rest and its final energy is

$$E_{\mathrm{sys},f} = Mc^2.$$

Equating the initial and final energies of the system shows that the mass M of the conglomerate particle is

$$M = \frac{2m}{\sqrt{1 - v^2/c^2}}.$$

Thus the particle mass has increased by an amount

$$\Delta m = M - 2m = \frac{2m}{\sqrt{1 - v^2/c^2}} - 2m = 2m\left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right).$$

The origin of this mass increase may be found by comparing the initial and final values of the kinetic energy. The initial kinetic energy of the system is

$$K_{\mathrm{sys},i} = 2mc^2\left(\frac{1}{\sqrt{1 - v^2/c^2}} - 1\right)$$

and the final kinetic energy $K_{\mathrm{sys},f} = 0$. Dividing the kinetic energy lost in this inelastic collision by c^2 equals the particle mass increase, Δm.

The Derivation of Relativistic Momentum (Eq. 4.44)

To justify Eq. (4.44) for the relativistic momentum, we will consider a glancing elastic collision between two identical particles of mass m. This collision will be observed from three carefully chosen inertial reference frames, as shown in Fig. 4.11. When measured in an inertial reference frame S'', the two particles A and B have velocities and momenta that are equal in magnitude and opposite in direction, both before and after the collision. As a result, the total momentum must be zero both before and after the collision; momentum is conserved. This collision can also be measured from two other reference frames, S and S'. From Fig. 4.11, if S moves in the negative x''-direction with a velocity equal to the x''-component of particle A in S'', then as measured from frame S, the velocity of particle A has only a y-component. Similarly, if S' moves in the positive x''-direction with a velocity equal to the x''-component of particle B in S'', then as measured from frame S', the velocity of particle B has only a y-component. Actually, the figures for frames S and S' would be *identical* if the figures for one of these frames were rotated by 180° and the A and B labels were reversed. This means that the change in the y-component of particle A's momentum as measured in frame S is the same as the change in the y'-component of particle B's momentum as measured in the frame S', except for a change in sign (due to the 180° rotation): $\Delta p_{A,y} = -\Delta p'_{B,y}$. On the other hand, momentum must be conserved in frames S and S', just as it is in frame S''. This means that, measured in frame S', the sum of the changes in the y'-components of particle A's and B's momenta must be zero: $\Delta p'_{A,y} + \Delta p'_{B,y} = 0$. Combining these results gives

$$\Delta p'_{A,y} = \Delta p_{A,y}. \tag{4.49}$$

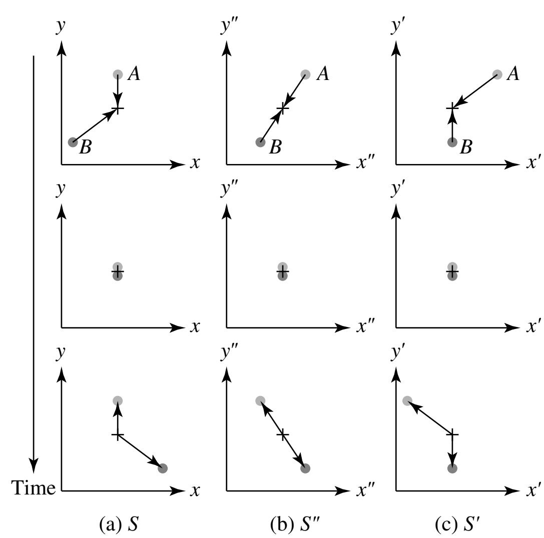

FIGURE 4.11 An elastic collision measured in frames (a) S, (b) S'', and (c) S'. As observed from frame S'', frame S moves in the negative x''-direction, along with particle A, and frame S' moves in the positive x''-direction, along with particle B. For each reference frame, a vertical sequence of three figures shows the situation before (top), during, and after the collision.

So far, the argument has been independent of a specific formula for the relativistic momentum vector $\mathbf{p}$. Let's assume that the relativistic momentum vector has the form $\mathbf{p} = fm\mathbf{v}$, where f is a relativistic factor that depends on the *magnitude* of the particle's velocity, but not its direction. As the particle's speed $v \to 0$, it is required that the factor $f \to 1$ to obtain agreement with the Newtonian result.[9]

A second assumption allows the relativistic factor f to be determined: The y- and y'-components of each particle's velocity are chosen to be arbitrarily small compared to the speed of light. Thus the y- and y'-components of particle A's velocity in frames S and S' are extremely small, and the x'-component of particle A's velocity in frame S' is taken to be relativistic. Since

$$v'_A = \sqrt{v'^2_{A,x} + v'^2_{A,y}} \approx c$$

in frame S', the relativistic factor f' for particle A in frame S' is not equal to 1, whereas in frame S, f is arbitrarily close to unity. If $v_{A,y}$ is the final y-component of particle A's velocity, and similarly for $v'_{A,y}$, then Eq. (4.49) becomes

$$2f' m v'_{A,y} = 2m v_{A,y}. \tag{4.50}$$

The relative velocity u of frames S and S' is needed to relate $v'_{A,y}$ and $v_{A,y}$ using Eq. (4.41). Because $v_{A,x} = 0$ in frame S, Eq. (4.40) shows that $u = -v'_{A,x}$; that is, the relative velocity u of frame S' relative to frame S is just the negative of the x'-component of particle A's velocity in frame S'. Furthermore, because the y'-component of particle A's velocity is arbitrarily small, we can set $v'_{A,x} = v'_A$, the magnitude of particle A's velocity as measured in frame S', and so use $u = -v'_A$. Inserting this into Eq. (4.41) with $v_{A,x} = 0$ gives

$$v'_{A,y} = v_{A,y}\sqrt{1 - v'^2_A/c^2}.$$

Finally, inserting this relation between $v'_{A,y}$ and $v_{A,y}$ into Eq. (4.50) and canceling terms reveals the relativistic factor f to be

$$f = \frac{1}{\sqrt{1 - v'^2_A/c^2}},$$

as measured in frame S'. Dropping the prime superscript and the A subscript (which merely identify the reference frame and particle involved) gives

$$f = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

The formula for the **relativistic momentum vector** $\mathbf{p} = fm\mathbf{v}$ is thus

$$\mathbf{p} = \frac{m\mathbf{v}}{\sqrt{1 - v^2/c^2}} = \gamma m\mathbf{v}.$$

[9]There is no requirement that relativistic formulas appear similar to their low-speed Newtonian counterparts (cf. Eq. 4.45). However, this simple argument produces the correct result.

SUGGESTED READING

General

French, A. P. (ed.), *Einstein: A Centenary Volume*, Harvard University Press, Cambridge, MA, 1979.

Gardner, Martin, *The Relativity Explosion*, Vintage Books, New York, 1976.

Technical

Becker, Robert H. et al., "Evidence for Reionization at $Z \sim 6$: Detection of a Gunn–Peterson Trough in a $Z = 6.28$ Quasar," *The Astronomical Journal*, preprint, 2001.

Bregman, Joel N. et al., "Multifrequency Observations of the Optically Violent Variable Quasar 3C 446," *The Astrophysical Journal*, *331*, 746, 1988.

Frisch, David H., and Smith, James H., "Measurement of the Relativistic Time Dilation Using μ-Mesons," *American Journal of Physics*, *31*, 342, 1963.

Hafele, J. C., and Keating, Richard E., "Around-the-World Atomic Clocks: Predicted Relativistic Time Gains," *Science*, *177*, 166, 1972a.

Hafele, J. C., and Keating, Richard E., "Around-the-World Atomic Clocks: Observed Relativistic Time Gains," *Science*, *177*, 168, 1972b.

McCarthy, Patrick J. et al., "Serendipitous Discovery of a Redshift 4.4 QSO," *The Astrophysical Journal Letters*, *328*, L29, 1988.

Sloan Digital Sky Survey, `http://www.sdss.org`

Resnick, Robert, and Halliday, David, *Basic Concepts in Relativity and Early Quantum Theory*, Second Edition, John Wiley and Sons, New York, 1985.

Taylor, Edwin F., and Wheeler, John A., *Spacetime Physics*, Second Edition, W. H. Freeman, San Francisco, 1992.

PROBLEMS

4.1 Use Eqs. (4.14) and (4.15) to derive the Lorentz transformation equations from Eqs. (4.10–4.13).

4.2 Because there is no such thing as absolute simultaneity, two observers in relative motion may disagree on which of two events A and B occurred first. Suppose, however, that an observer in reference frame S measures that event A occurred first and *caused* event B. For example, event A might be pushing a light switch, and event B might be a light bulb turning on. Prove that an observer in another frame S' cannot measure event B (the effect) occurring before event A (the cause). The temporal order of cause and effect is preserved by the Lorentz transformation equations. *Hint:* For event A to cause event B, information must have traveled from A to B, and the fastest that *anything* can travel is the speed of light.

4.3 Consider the special *light clock* shown in Fig. 4.12. The light clock is at rest in frame S' and consists of two perfectly reflecting mirrors separated by a vertical distance d. As measured by an observer in frame S', a light pulse bounces vertically back and forth between the two mirrors; the time interval between the pulse leaving and subsequently returning to the bottom mirror is $\Delta t'$. However, an observer in frame S sees a moving clock and determines that the time interval

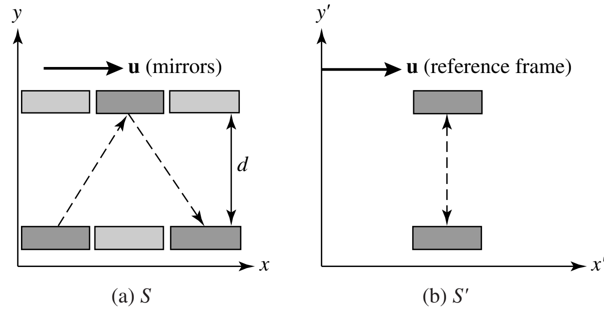

FIGURE 4.12 (a) A light clock that is moving in frame S, and (b) at rest in frame S'.

between the light pulse leaving and returning to the bottom mirror is Δt. Use the fact that both observers must measure that the light pulse moves with speed c, plus some simple geometry, to derive the time-dilation equation (4.27).

4.4 A rod moving relative to an observer is measured to have its length L_{moving} contracted to one-half of its length when measured at rest. Find the value of u/c for the rod's rest frame relative to the observer's frame of reference.

4.5 An observer P stands on a train station platform as a high-speed train passes by at $u/c = 0.8$. The observer P, who measures the platform to be 60 m long, notices that the front and back ends of the train line up exactly with the ends of the platform at the same time.

(a) How long does it take the train to pass P as he stands on the platform, as measured by his watch?

(b) According to a rider T on the train, how long is the train?

(c) According to a rider T on the train, what is the length of the train station platform?

(d) According to a rider T on the train, how much time does it take for the train to pass observer P standing on the train station platform?

(e) According to a rider T on the train, the ends of the train will *not* simultaneously line up with the ends of the platform. What time interval does T measure between when the front end of the train lines up with the front end of the platform, and when the back end of the train lines up with the back end of the platform?

4.6 An astronaut in a starship travels to α Centauri, a distance of approximately 4 ly as measured from Earth, at a speed of $u/c = 0.8$.

(a) How long does the trip to α Centauri take, as measured by a clock on Earth?

(b) How long does the trip to α Centauri take, as measured by the starship pilot?

(c) What is the distance between Earth and α Centauri, as measured by the starship pilot?

(d) A radio signal is sent from Earth to the starship every 6 months, as measured by a clock on Earth. What is the time interval between reception of one of these signals and reception of the next signal aboard the starship?

(e) A radio signal is sent from the starship to Earth every 6 months, as measured by a clock aboard the starship. What is the time interval between reception of one of these signals and reception of the next signal on Earth?

(f) If the wavelength of the radio signal sent from Earth is $\lambda = 15$ cm, to what wavelength must the starship's receiver be tuned?

4.7 Upon reaching α Centauri, the starship in Problem 4.6 immediately reverses direction and travels back to Earth at a speed of $u/c = 0.8$. (Assume that the turnaround itself takes *zero* time.) Both Earth and the starship continue to emit radio signals at 6-month intervals, as measured by their respective clocks. Make a table for the entire trip showing at what times Earth receives the signals from the starship. Do the same for the times when the starship receives the signals from Earth. Thus an Earth observer and the starship pilot will agree that the pilot has aged 4 years less than the Earth observer during the round-trip voyage to α Centauri.

4.8 In its rest frame, quasar Q2203+29 produces a hydrogen emission line of wavelength 121.6 nm. Astronomers on Earth measure a wavelength of 656.8 nm for this line. Determine the redshift parameter and the apparent speed of recession for this quasar. (For more information about this quasar, see McCarthy et al. 1988.)

4.9 Quasar 3C 446 is violently variable; its luminosity at optical wavelengths has been observed to change by a factor of 40 in as little as 10 days. Using the redshift parameter $z = 1.404$ measured for 3C 446, determine the time for the luminosity variation as measured in the quasar's rest frame. (For more details, see Bregman et al. 1988.)

4.10 Use the Lorentz transformation equations (4.16–4.19) to derive the velocity transformation equations (4.40–4.42).

4.11 The *spacetime interval*, Δs, between two events with coordinates

$$(x_1, y_1, z_1, t_1) \qquad \text{and} \qquad (x_2, y_2, z_2, t_2)$$

is defined by

$$(\Delta s)^2 \equiv (c\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2.$$

(a) Use the Lorentz transformation equations (4.16–4.19) to show that Δs has the same value in all reference frames. The spacetime interval is said to be *invariant* under a Lorentz transformation.

(b) If $(\Delta s)^2 > 0$, then the interval is *timelike*. Show that in this case,

$$\Delta\tau \equiv \frac{\Delta s}{c}$$

is the proper time between the two events. Assuming that $t_1 < t_2$, could the first event possibly have caused the second event?

(c) If $(\Delta s)^2 = 0$, then the interval is *lightlike* or *null*. Show that only light could have traveled between the two events. Could the first event possibly have caused the second event?

(d) If $(\Delta s)^2 < 0$, then the interval is *spacelike*. What is the physical significance of $\sqrt{-(\Delta s)^2}$? Could the first event possibly have caused the second event?

The concept of a spacetime interval will play a key role in the discussion of general relativity in Chapter 17.

4.12 General expressions for the components of a light ray's velocity as measured in reference frame S are

$$v_x = c \sin\theta \cos\phi$$

$$v_y = c \sin\theta \sin\phi$$

$$v_z = c \cos\theta,$$

where θ and ϕ are the angular coordinates in a spherical coordinate system.

(a) Show that

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2} = c.$$

(b) Use the velocity transformation equations to show that, as measured in reference frame S',

$$v' = \sqrt{v_x'^2 + v_y'^2 + v_z'^2} = c,$$

and so confirm that the speed of light has the constant value c in all frames of reference.

4.13 Starship A moves away from Earth with a speed of $v_A/c = 0.8$. Starship B moves away from Earth in the opposite direction with a speed of $v_B/c = 0.6$. What is the speed of starship A as measured by starship B? What is the speed of starship B as measured by starship A?

4.14 Use Newton's second law, $\mathbf{F} = d\mathbf{p}/dt$, and the formula for relativistic momentum, Eq. (4.44), to show that the acceleration vector $\mathbf{a} = d\mathbf{v}/dt$ produced by a force $\mathbf{F}$ acting on a particle of mass m is

$$\mathbf{a} = \frac{\mathbf{F}}{\gamma m} - \frac{\mathbf{v}}{\gamma m c^2}\,(\mathbf{F}\cdot\mathbf{v}),$$

where $\mathbf{F}\cdot\mathbf{v}$ is the vector dot product between the force $\mathbf{F}$ and the particle velocity $\mathbf{v}$. Thus the acceleration depends on the particle's velocity and is not in general in the same direction as the force.

4.15 Suppose a constant force of magnitude F acts on a particle of mass m initially at rest.

(a) Integrate the formula for the acceleration found in Problem 4.14 to show that the speed of the particle after time t is given by

$$\frac{v}{c} = \frac{(F/m)t}{\sqrt{(F/m)^2t^2 + c^2}}.$$

(b) Rearrange this equation to express the time t as a function of v/c. If the particle's initial acceleration at time $t = 0$ is $a = g = 9.80\ \mathrm{m\ s^{-2}}$, how much time is required for the particle to reach a speed of $v/c = 0.9$? $v/c = 0.99$? $v/c = 0.999$? $v/c = 0.9999$? $v/c = 1$?

4.16 Find the value of v/c when a particle's kinetic energy equals its rest energy.

4.17 Prove that in the low-speed Newtonian limit of $v/c \ll 1$, Eq. (4.45) does reduce to the familiar form $K = \frac{1}{2}mv^2$.

4.18 Show that the relativistic kinetic energy of a particle can be written as

$$K = \frac{p^2}{(1+\gamma)m},$$

where p is the magnitude of the particle's relativistic momentum. This demonstrates that in the low-speed Newtonian limit of $v/c \ll 1$, $K = p^2/2m$ (as expected).

4.19 Derive Eq. (4.48).

CHAPTER 5 The Interaction of Light and Matter

5.1 ■ SPECTRAL LINES

In 1835 a French philosopher, Auguste Comte (1798–1857), considered the limits of human knowledge. In his book *Positive Philosophy*, Comte wrote of the stars, "We see how we may determine their forms, their distances, their bulk, their motions, but we can never know anything of their chemical or mineralogical structure." Thirty-three years earlier, however, William Wollaston (1766–1828), like Newton before him, passed sunlight through a prism to produce a rainbow-like spectrum. He discovered that a number of dark **spectral lines** were superimposed on the continuous spectrum where the Sun's light had been absorbed at certain discrete wavelengths. By 1814, the German optician Joseph von Fraunhofer (1787–1826) had cataloged 475 of these dark lines (today called **Fraunhofer lines**) in the solar spectrum. While measuring the wavelengths of these lines, Fraunhofer made the first observation capable of proving Comte wrong. Fraunhofer determined that the wavelength of one prominent dark line in the Sun's spectrum corresponds to the wavelength of the yellow light emitted when salt is sprinkled in a flame. The new science of *spectroscopy* was born with the identification of this sodium line.

Kirchhoff's Laws

The foundations of spectroscopy were established by Robert Bunsen (1811–1899), a German chemist, and by Gustav Kirchhoff (1824–1887), a Prussian theoretical physicist. Bunsen's burner produced a colorless flame that was ideal for studying the spectra of heated substances. He and Kirchhoff then designed a *spectroscope* that passed the light of a flame spectrum through a prism to be analyzed. The wavelengths of light absorbed and emitted by an element were found to be the same; Kirchhoff determined that 70 dark lines in the solar spectrum correspond to 70 bright lines emitted by iron vapor. In 1860 Kirchhoff and Bunsen published their classic work *Chemical Analysis by Spectral Observations*, in which they developed the idea that every element produces its own pattern of spectral lines and thus may be identified by its unique spectral line "fingerprint." Kirchhoff summarized the production of spectral lines in three laws, which are now known as **Kirchhoff's laws**:

- A hot, dense gas or hot solid object produces a continuous spectrum with no dark spectral lines.[1]
- A hot, diffuse gas produces bright spectral lines (**emission lines**).
- A cool, diffuse gas in front of a source of a continuous spectrum produces dark spectral lines (**absorption lines**) in the continuous spectrum.

Applications of Stellar Spectra Data

An immediate application of these results was the identification of elements found in the Sun and other stars. A new element previously unknown on Earth, *helium*,[2] was discovered spectroscopically on the Sun in 1868; it was not found on Earth until 1895. Figure 5.1 shows the visible portion of the solar spectrum, and Table 5.1 lists some of the elements responsible for producing the dark absorption lines.

Another rich line of investigation was pursued by measuring the Doppler shifts of spectral lines. For individual stars, $v_r \ll c$, and so the *low-speed approximation* of Eq. (4.39),

$$\frac{\lambda_{\text{obs}} - \lambda_{\text{rest}}}{\lambda_{\text{rest}}} = \frac{\Delta\lambda}{\lambda_{\text{rest}}} = \frac{v_r}{c}, \tag{5.1}$$

can be utilized to determine their radial velocities. By 1887 the radial velocities of Sirius, Procyon, Rigel, and Arcturus had been measured with an accuracy of a few kilometers per second.

Example 5.1.1. The rest wavelength λ_{rest} for an important spectral line of hydrogen (known as Hα) is 656.281 nm when measured in air. However, the wavelength of the Hα absorption line in the spectrum of the star Vega in the constellation Lyra is measured to be 656.251 nm at a ground-based telescope. Equation (5.1) shows that the radial velocity of Vega is

$$v_r = \frac{c\,(\lambda_{\text{obs}} - \lambda_{\text{rest}})}{\lambda_{\text{rest}}} = -13.9 \text{ km s}^{-1};$$

the minus sign means that Vega is approaching the Sun. Recall from Section 1.3, however, that stars also have a *proper motion*, μ, perpendicular to the line of sight. Vega's angular position in the sky changes by $\mu = 0.35077''\ \text{yr}^{-1}$. At a distance of $r = 7.76$ pc, this proper motion is related to the star's transverse velocity, v_θ, by Eq. (1.5). Expressing r in meters and μ in radians per second results in

$$v_\theta = r\mu = 12.9 \text{ km s}^{-1}.$$

[1]In the first of Kirchhoff's laws, "hot" actually means any temperature above 0 K. However, according to Wien's displacement law (Eq. 3.19), a temperature of several thousand degrees K is required for λ_{max} to fall in the visible portion of the electromagnetic spectrum. As will be discussed in Chapter 9, it is the opacity or *optical depth* of the gas that is responsible for the continuous blackbody spectrum.

[2]The name *helium* comes from *Helios*, a Greek Sun god.

This transverse velocity is comparable to Vega's radial velocity. Vega's speed through space relative to the Sun is thus

$$v = \sqrt{v_r^2 + v_\theta^2} = 19.0 \text{ km s}^{-1}.$$

The average speed of stars in the solar neighborhood is about 25 km s^{-1}. In reality, the measurement of a star's radial velocity is complicated by the 29.8 km s^{-1} motion of Earth around the Sun, which causes the observed wavelength λ_{obs} of a spectral line to vary sinusoidally over the course of a year. This effect of Earth's speed may be easily compensated for by subtracting the component of Earth's orbital velocity along the line of sight from the star's measured radial velocity.

Spectrographs

Modern methods can measure radial velocities with an accuracy of better than ± 3 m s^{-1}! Today astronomers use *spectrographs* to measure the spectra of stars and galaxies; see Fig. 5.2.[3] After passing through a narrow slit, the starlight is collimated by a mirror and directed onto a *diffraction grating*. A diffraction grating is a piece of glass onto which narrow, closely spaced lines have been evenly ruled (typically several thousand lines per millimeter); the grating may be made to transmit the light (a *transmission grating*) or reflect the light (a *reflection grating*). In either case, the grating acts like a long series of neighboring double slits. Different wavelengths of light have their maxima occurring at different angles θ given by Eq. (3.11):

$$d \sin\theta = n\lambda \qquad (n = 0, 1, 2, \ldots),$$

where d is the distance between adjacent lines of the grating, n is the order of the spectrum, and θ is measured from the line normal (or perpendicular) to the grating. ($n = 0$ corresponds to $\theta = 0$ for all wavelengths, so the light is not dispersed into a spectrum in this case.) The spectrum is then focused onto a photographic plate or electronic detector for recording.

The ability of a spectrograph to resolve two closely spaced wavelengths separated by an amount $\Delta\lambda$ depends on the order of the spectrum, n, and the total number of lines of the grating that are illuminated, N. The smallest difference in wavelength that the grating can resolve is

$$\Delta\lambda = \frac{\lambda}{nN}, \tag{5.2}$$

where λ is either of the closely spaced wavelengths being measured. The ratio $\lambda/\Delta\lambda$ is the **resolving power** of the grating.[4]

[3]As we will discuss in Chapter 7, measuring the radial velocities of stars in binary star systems allows the *masses* of the stars to be determined. The same methods have now been used to detect numerous extrasolar planets.
[4]In some cases, the resolving power of a spectrograph may be determined by other factors—for example, the slit width.

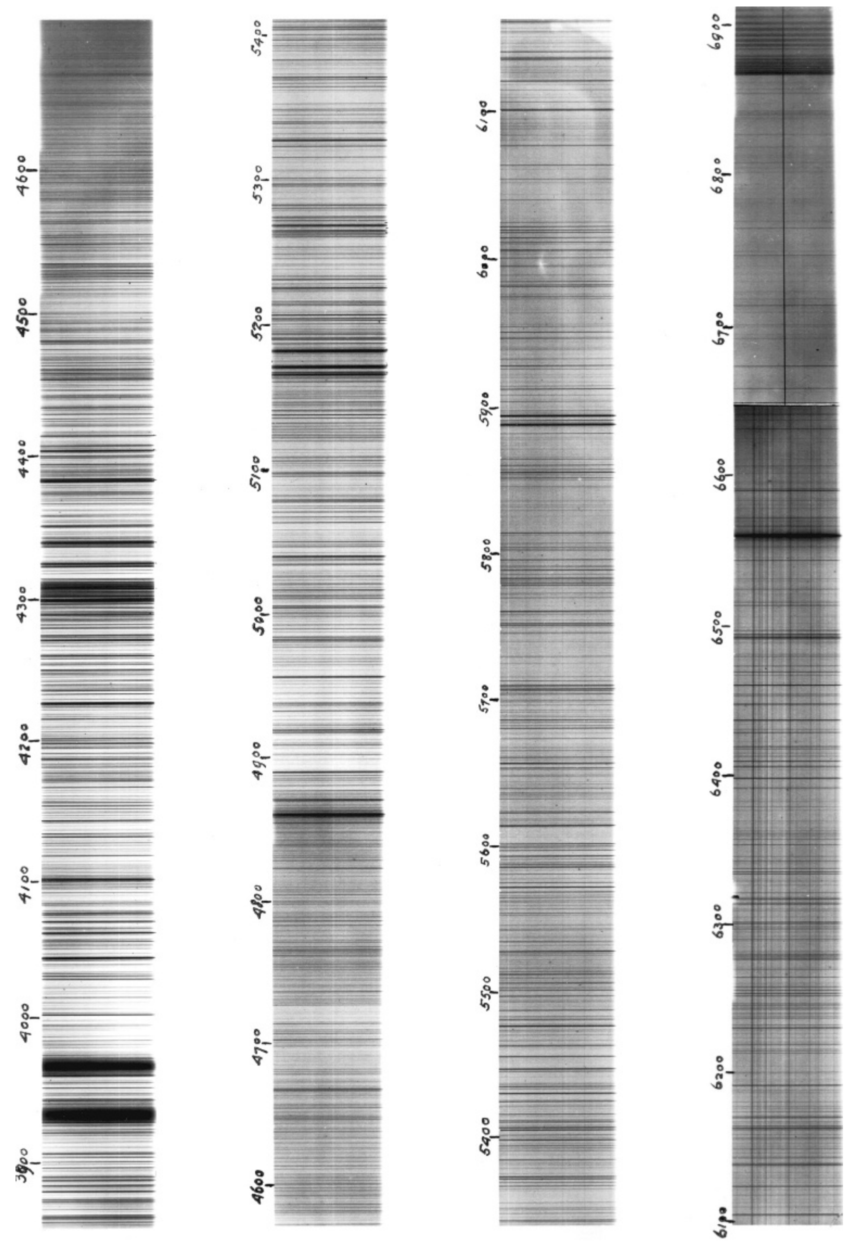

FIGURE 5.1 The solar spectrum with Fraunhofer lines. Note that the wavelengths are expressed in angstroms (1 Å = 0.1 nm), a commonly used wavelength unit in astronomy. Modern depictions of spectra are typically shown as plots of flux as a function of wavelength; see, for instance, Figs. 8.4 and 8.5. (Courtesy of The Observatories of the Carnegie Institution of Washington.)

TABLE 5.1 Wavelengths of some of the stronger Fraunhofer lines measured in air near sea level. The atomic notation is explained in Section 8.1, and the equivalent width of a spectral line is defined in Section 9.5. The difference in wavelengths of spectral lines when measured in air versus in vacuum are discussed in Example 5.3.1. (Data from Lang, *Astrophysical Formulae*, Third Edition, Springer, New York, 1999.)

Wavelength (nm)	Name	Atom	Equivalent Width (nm)
385.992		Fe I	0.155
388.905		H_8	0.235
393.368	K	Ca II	2.025
396.849	H	Ca II	1.547
404.582		Fe I	0.117
410.175	h, Hδ	H I	0.313
422.674	g	Ca I	0.148
434.048	G$'$, Hγ	H I	0.286
438.356	d	Fe I	0.101
486.134	F, Hβ	H I	0.368
516.733	b_4	Mg I	0.065
517.270	b_2	Mg I	0.126
518.362	b_1	Mg I	0.158
588.997	D_2	Na I	0.075
589.594	D_1	Na I	0.056
656.281	C, Hα	H I	0.402

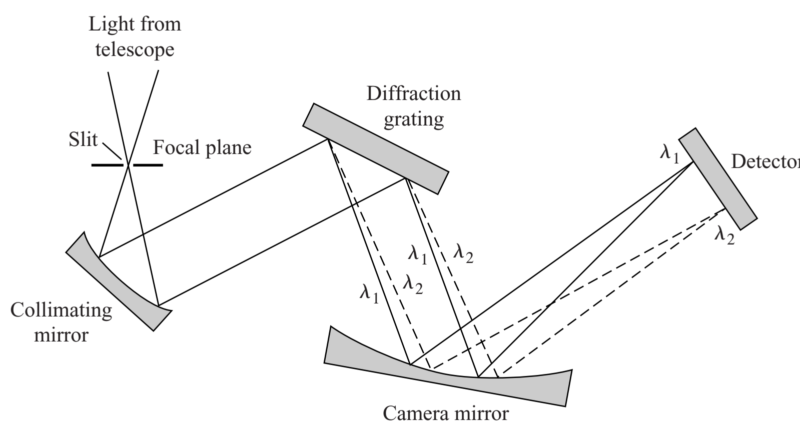

FIGURE 5.2 Spectrograph.

Astronomers recognized the great potential for uncovering the secrets of the stars in the empirical rules that had been obtained for the spectrum of light: Wien's law, the Stefan–Boltzmann equation, Kirchhoff's laws, and the new science of spectroscopy. By 1880 Gustav Wiedemann (1826–1899) found that a detailed investigation of the Fraunhofer lines could reveal the temperature, pressure, and density of the layer of the Sun's atmosphere that produces the lines. The splitting of spectral lines by a magnetic field was discovered by Pieter Zeeman (1865–1943) of the Netherlands in 1897, raising the possibility of measuring stellar magnetic fields. But a serious problem blocked further progress: However impressive, these results lacked the solid theoretical foundation required for the interpretation of stellar spectra. For example, the absorption lines produced by hydrogen are much stronger for Vega than for the Sun. Does this mean that Vega's composition contains significantly more hydrogen than the Sun's? The answer is no, but how can this information be gleaned from the dark absorption lines of a stellar spectrum recorded on a photographic plate? The answer required a new understanding of the nature of light itself.

5.2 ■ PHOTONS

Despite Heinrich Hertz's absolute certainty in the wave nature of light, the solution to the riddle of the continuous spectrum of blackbody radiation led to a complementary description, and ultimately to new conceptions of matter and energy. Planck's constant h (see Section 3.5) is the basis of the modern description of matter and energy known as **quantum mechanics**. Today h is recognized as a fundamental constant of nature, like the speed of light c and the universal gravitational constant G. Although Planck himself was uncomfortable with the implications of his discovery of energy quantization, quantum theory was to develop into what is today a spectacularly successful description of the physical world. The next step forward was taken by Einstein, who convincingly demonstrated the reality of Planck's quantum bundles of energy.

The Photoelectric Effect

When light shines on a metal surface, electrons are ejected from the surface, a result called the **photoelectric effect**. The electrons are emitted with a range of energies, but those originating closest to the surface have the maximum kinetic energy, $K_{\max}$. A surprising feature of the photoelectric effect is that the value of $K_{\max}$ does *not* depend on the brightness of the light shining on the metal. Increasing the intensity of a monochromatic light source will eject *more* electrons but will not increase their maximum kinetic energy. Instead, $K_{\max}$ varies with the *frequency* of the light illuminating the metal surface. In fact, each metal has a characteristic *cutoff frequency* ν_c and a corresponding *cutoff wavelength* $\lambda_c = c/\nu_c$; electrons will be emitted only if the frequency ν of the light satisfies $\nu > \nu_c$ (or the wavelength satisfies $\lambda < \lambda_c$). This puzzling frequency dependence is nowhere to be found in Maxwell's classic description of electromagnetic waves. Equation (3.12) for the Poynting vector admits no role for the frequency in describing the energy carried by a light wave.

Einstein's bold solution was to take seriously Planck's assumption of the quantized energy of electromagnetic waves. According to Einstein's explanation of the photoelectric effect, the light striking the metal surface consists of a stream of massless particles called

photons.[5] The energy of a single photon of frequency ν and wavelength λ is just the energy of Planck's quantum of energy:

$$E_{\text{photon}} = h\nu = \frac{hc}{\lambda}. \tag{5.3}$$

Example 5.2.1. The energy of a single photon of visible light is small by everyday standards. For red light of wavelength $\lambda = 700$ nm, the energy of a single photon is

$$E_{\text{photon}} = \frac{hc}{\lambda} \simeq \frac{1240 \text{ eV nm}}{700 \text{ nm}} = 1.77 \text{ eV}.$$

Here, the product hc has been expressed in the convenient units of (electron volts) $\times$ (nanometers); recall that 1 eV $= 1.602 \times 10^{-19}$ J. For a single photon of blue light with $\lambda = 400$ nm,

$$E_{\text{photon}} = \frac{hc}{\lambda} \simeq \frac{1240 \text{ eV nm}}{400 \text{ nm}} = 3.10 \text{ eV}.$$

How many visible photons ($\lambda = 500$ nm) are emitted each second by a 100-W light bulb (assuming that it is monochromatic)? The energy of each photon is

$$E_{\text{photon}} = \frac{hc}{\lambda} \simeq \frac{1240 \text{ eV nm}}{500 \text{ nm}} = 2.48 \text{ eV} = 3.97 \times 10^{-19} \text{ J}.$$

This means that the 100-W light bulb emits 2.52×10^{20} photons per second. As this huge number illustrates, with so many photons nature does not appear "grainy." We see the world as a continuum of light, illuminated by a flood of photons.

Einstein reasoned that when a photon strikes the metal surface in the photoelectric effect, its energy may be absorbed by a single electron. The electron uses the photon's energy to overcome the binding energy of the metal and so escape from the surface. If the *minimum* binding energy of electrons in a metal (called the **work function** of the metal, usually a few eV) is ϕ, then the maximum kinetic energy of the ejected electrons is

$$\boxed{K_{\text{max}} = E_{\text{photon}} - \phi = h\nu - \phi = \frac{hc}{\lambda} - \phi.} \tag{5.4}$$

Setting $K_{\text{max}} = 0$, the cutoff frequency and wavelength for a metal are seen to be $\nu_c = \phi/h$ and $\lambda_c = hc/\phi$, respectively.

The photoelectric effect established the reality of Planck's quanta. Albert Einstein was awarded the 1921 Nobel Prize, not for his theories of special and general relativity, but "for his services to theoretical physics, and especially for his discovery of the law of the

[5] Only a massless particle can move with the speed of light, since a massive particle would have infinite energy; see Eq. (4.45). The term *photon* was first used in 1926 by the physicist G. N. Lewis (1875–1946).

photoelectric effect."[6] Today astronomers take advantage of the quantum nature of light in various instruments and detectors, such as the CCDs (charge-coupled devices) that will be described in Section 6.2.

The Compton Effect

In 1922, the American physicist Arthur Holly Compton (1892–1962) provided the most convincing evidence that light does in fact manifest its particle-like nature when interacting with matter. Compton measured the change in the wavelength of X-ray photons as they were scattered by free electrons. Because photons are massless particles that move at the speed of light, the relativistic energy equation, Eq. (4.48) (with mass $m = 0$ for photons), shows that the energy of a photon is related to its momentum p by

$$\boxed{E_{\text{photon}} = h\nu = \frac{hc}{\lambda} = pc.} \tag{5.5}$$

Compton considered the "collision" between a photon and a free electron, initially at rest. As shown in Fig. 5.3, the electron is scattered in the direction ϕ and the photon is scattered by an angle θ. Because the photon has lost energy to the electron, the wavelength of the photon has increased.

In this collision, both (relativistic) momentum and energy are conserved. It is left as an exercise to show that the final wavelength of the photon, λ_f, is greater than its initial

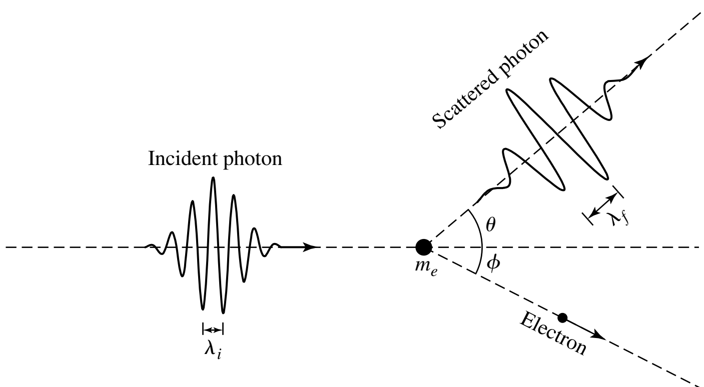

FIGURE 5.3 The Compton effect: The scattering of a photon by a free electron. θ and ϕ are the scattering angles of the photon and electron, respectively.

[6]Partly in recognition of his determination of an accurate value of Planck's constant h using Eq. (5.4), the American physicist Robert A. Millikan (1868–1953) also received a Nobel Prize (1923) for his work on the photoelectric effect.

wavelength, λ_i, by an amount

$$\boxed{\Delta\lambda = \lambda_f - \lambda_i = \frac{h}{m_e c}(1 - \cos\theta),} \tag{5.6}$$

where m_e is the mass of the electron. Today, this change in wavelength is known as the **Compton effect**. The term $h/m_e c$ in Eq. (5.6), called the **Compton wavelength**, λ_C, is the characteristic change in the wavelength of the scattered photon and has the value $\lambda_C =$ 0.00243 nm, 30 times smaller than the wavelength of the X-ray photons used by Compton. Compton's experimental verification of this formula provided convincing evidence that photons are indeed massless particles that nonetheless carry momentum, as described by Eq. (5.5). This is the physical basis for the force exerted by radiation upon matter, which we discussed in Section 3.3 in terms of radiation pressure.

5.3 ■ THE BOHR MODEL OF THE ATOM

The pioneering work of Planck, Einstein, and others at the beginning of the twentieth century revealed the **wave–particle duality** of light. Light exhibits its wave properties as it *propagates* through space, as demonstrated by its double-slit interference pattern. On the other hand, light manifests its particle nature when it *interacts* with matter, as in the photoelectric and Compton effects. Planck's formula describing the energy distribution of blackbody radiation explained many of the features of the continuous spectrum of light emitted by stars. But what physical process was responsible for the dark absorption lines scattered throughout the continuous spectrum of a star, or for the bright emission lines produced by a hot, diffuse gas in the laboratory?

The Structure of the Atom

In the very last years of the nineteenth century, Joseph John Thomson (1856–1940) discovered the **electron** while working at Cambridge University's Cavendish Laboratory. Because bulk matter is electrically neutral, atoms were deduced to consist of negatively charged electrons and an equal positive charge of uncertain distribution. Ernest Rutherford (1871–1937) of New Zealand, working at England's University of Manchester, discovered in 1911 that an atom's positive charge was concentrated in a tiny, massive nucleus. Rutherford directed high-speed alpha particles (now known to be helium nuclei) onto thin metal foils. He was amazed to observe that a few of the alpha particles were bounced backward by the foils instead of plowing through them with only a slight deviation. Rutherford later wrote: "It was quite the most incredible event that has ever happened to me in my life. It was almost as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you." Such an event could occur only as the result of a single collision of the alpha particle with a minute, massive, positively charged nucleus. Rutherford calculated that the radius of the nucleus was 10,000 times smaller than the radius of the atom itself, showing that ordinary matter is mostly empty space! He established that an electrically neutral atom consists of Z electrons (where Z is an integer), with Z positive elementary charges confined to the nucleus. Rutherford coined the term **proton** to refer to the nucleus of the hydrogen atom ($Z = 1$), 1836 times more massive than the electron. But how were these charges arranged?

The Wavelengths of Hydrogen

The experimental data were abundant. The wavelengths of 14 spectral lines of hydrogen had been precisely determined. Those in the visible region of the electromagnetic spectrum are 656.3 nm (red, Hα), 486.1 nm (turquoise, Hβ), 434.0 nm (blue, Hγ), and 410.2 nm (violet, Hδ). In 1885 a Swiss school teacher, Johann Balmer (1825–1898), had found, by trial and error, a formula to reproduce the wavelengths of these spectral lines of hydrogen, today called the **Balmer series** or **Balmer lines**:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{4} - \frac{1}{n^2} \right), \tag{5.7}$$

where $n = 3, 4, 5, \ldots$, and $R_H = 1.09677583 \times 10^7 \pm 1.3\ \mathrm{m^{-1}}$ is the experimentally determined Rydberg constant for hydrogen.[7] Balmer's formula was very accurate, to within a fraction of a percent. Inserting $n = 3$ gives the wavelength of the Hα Balmer line, $n = 4$ gives Hβ, and so on. Furthermore, Balmer realized that since $2^2 = 4$, his formula could be generalized to

$$\frac{1}{\lambda} = R_H \left(\frac{1}{m^2} - \frac{1}{n^2} \right), \tag{5.8}$$

with $m < n$ (both integers). Many nonvisible spectral lines of hydrogen were found later, just as Balmer had predicted. Today, the lines corresponding to $m = 1$ are called *Lyman lines*. The Lyman series of lines is found in the ultraviolet region of the electromagnetic spectrum. Similarly, inserting $m = 3$ into Eq. (5.8) produces the wavelengths of the *Paschen* series of lines, which lie entirely in the infrared portion of the spectrum. The wavelengths of important selected hydrogen lines are given in Table 5.2.

Yet all of this was sheer numerology, with no foundation in the physics of the day. Physicists were frustrated by their inability to construct a model of even this simplest of atoms. A planetary model of the hydrogen atom, consisting of a central proton and one electron held together by their mutual electrical attraction, should have been most amenable to analysis. However, a model consisting of a single electron and proton moving around their common center of mass suffers from a basic instability. According to Maxwell's equations of electricity and magnetism, an accelerating electric charge emits electromagnetic radiation. The orbiting electron should thus lose energy by emitting light with a continuously increasing frequency (the orbital frequency) as it spirals down into the nucleus. This theoretical prediction of a continuous spectrum disagreed with the discrete emission lines actually observed. Even worse was the calculated timescale: The electron should plunge into the nucleus in only 10^{-8} s. Obviously, matter is stable over much longer periods of time!

[7] R_H is named in honor of Johannes Rydberg (1854–1919), a Swedish spectroscopist.

TABLE 5.2 The wavelengths of selected hydrogen spectral lines measured in vacuo for the Lyman lines and in air for the Balmer and Paschen lines. (Based on Cox, (ed.), *Allen's Astrophysical Quantities*, Fourth Edition, Springer, New York, 2000.)

Series Name	Symbol	Transition	Wavelength (nm)
Lyman	Lyα	$2 \leftrightarrow 1$	121.567
	Lyβ	$3 \leftrightarrow 1$	102.572
	Lyγ	$4 \leftrightarrow 1$	97.254
	Ly$_{\text{limit}}$	$\infty \leftrightarrow 1$	91.18
Balmer	Hα	$3 \leftrightarrow 2$	656.281
	Hβ	$4 \leftrightarrow 2$	486.134
	Hγ	$5 \leftrightarrow 2$	434.048
	Hδ	$6 \leftrightarrow 2$	410.175
	Hϵ	$7 \leftrightarrow 2$	397.007
	H$_8$	$8 \leftrightarrow 2$	388.905
	H$_{\text{limit}}$	$\infty \leftrightarrow 2$	364.6
Paschen	Paα	$4 \leftrightarrow 3$	1875.10
	Paβ	$5 \leftrightarrow 3$	1281.81
	Paγ	$6 \leftrightarrow 3$	1093.81
	Pa$_{\text{limit}}$	$\infty \leftrightarrow 3$	820.4

Bohr's Semiclassical Atom

Theoretical physicists hoped that the answer might be found among the new ideas of photons and quantized energy. A Danish physicist, Niels Bohr (1885–1962; see Fig. 5.4) came to the rescue in 1913 with a daring proposal. The dimensions of Planck's constant, J $\times$ s, are equivalent to kg $\times$ m s^{-1} $\times$ m, the units of angular momentum. Perhaps the angular momentum of the orbiting electron was quantized. This quantization had been previously introduced into atomic models by the British astronomer J. W. Nicholson. Although Bohr knew that Nicholson's models were flawed, he recognized the possible significance of the quantization of angular momentum. Just as an electromagnetic wave of frequency ν could have the energy of only an integral number of quanta, $E = nh\nu$, suppose that the value of the angular momentum of the hydrogen atom could assume only integral multiples of Planck's constant divided by 2π: $L = nh/2\pi = n\hbar$.[8] Bohr hypothesized that in orbits with precisely these allowed values of the angular momentum, the electron would be stable and would not radiate *in spite of its centripetal acceleration*. What would be the result of such a bold departure from classical physics?

To analyze the mechanical motion of the atomic electron–proton system, we start with the mathematical description of their electrical attraction given by **Coulomb's law**. For two charges q_1 and q_2 separated by a distance r, the electric force on charge 2 due to charge 1 has the familiar form

$$\boxed{\mathbf{F} = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\,\hat{\mathbf{r}},} \tag{5.9}$$

[8]The quantity $\hbar \equiv h/2\pi = 1.054571596 \times 10^{-34}$ J s and is pronounced "h-bar."

FIGURE 5.4 Niels Bohr (1885–1962). (Courtesy of The Niels Bohr Archive, Copenhagen.)

where $\epsilon_0 = 8.854187817\ldots \times 10^{-12}$ F m^{-1} is the *permittivity of free space*[9] and $\hat{\mathbf{r}}$ is a unit vector directed from charge 1 toward charge 2.

Consider an electron of mass m_e and charge $-e$ and a proton of mass m_p and charge $+e$ in circular orbits around their common center of mass, under the influence of their mutual electrical attraction, e being the *fundamental charge*, $e = 1.602176462 \times 10^{-19}$ C. Recall from Section 2.3 that this two-body problem may be treated as an equivalent one-body problem by using the reduced mass

$$\mu = \frac{m_e m_p}{m_e + m_p} = \frac{(m_e)(1836.15266\, m_e)}{m_e + 1836.15266\, m_e} = 0.999455679\, m_e$$

and the total mass

$$M = m_e + m_p = m_e + 1836.15266\, m_e = 1837.15266\, m_e = 1.0005446\, m_p$$

of the system. Since $M \simeq m_p$ and $\mu \simeq m_e$, the hydrogen atom may be thought of as being composed of a proton of mass M that is at rest and an electron of mass μ that follows a circular orbit of radius r around the proton; see Fig. 5.5. The electrical attraction between the electron and the proton produces the electron's centripetal acceleration v^2/r, as described by Newton's second law:

$$\mathbf{F} = \mu \mathbf{a},$$

implying

$$\frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r^2}\,\hat{\mathbf{r}} = -\mu\,\frac{v^2}{r}\,\hat{\mathbf{r}},$$

[9]Formally, ϵ_0 is *defined* as $\epsilon_0 \equiv 1/\mu_0 c^2$, where $\mu_0 \equiv 4\pi \times 10^{-7}$ N A^{-2} is the *permeability of free space* and $c \equiv 2.99792458 \times 10^8$ m s^{-1} is the defined speed of light.

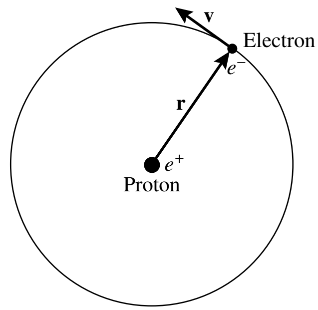

FIGURE 5.5 The Bohr model of the hydrogen atom.

or

$$-\frac{1}{4\pi\epsilon_0}\frac{e^2}{r^2}\,\hat{\mathbf{r}} = -\mu\,\frac{v^2}{r}\,\hat{\mathbf{r}}.$$

Canceling the minus sign and the unit vector $\hat{\mathbf{r}}$, this expression can be solved for the kinetic energy, $\frac{1}{2}\mu v^2$:

$$K = \frac{1}{2}\mu v^2 = \frac{1}{8\pi\epsilon_0}\frac{e^2}{r}. \tag{5.10}$$

Now the electrical potential energy U of the Bohr atom is[10]

$$U = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r} = -2K.$$

Thus the total energy $E = K + U$ of the atom is

$$E = K + U = K - 2K = -K = -\frac{1}{8\pi\epsilon_0}\frac{e^2}{r}. \tag{5.11}$$

Note that the relation between the kinetic, potential, and total energies is in accordance with the virial theorem for an inverse-square force, as discussed for gravity in Section 2.4, $E = \frac{1}{2}U = -K$. Because the kinetic energy must be positive, the total energy E is negative. This merely indicates that the electron and the proton are *bound*. To **ionize** the atom (that is, to remove the proton and electron to an infinite separation), an amount of energy of magnitude $|E|$ (or more) must be added to the atom.

Thus far the derivation has been completely classical in nature. At this point, however, we can use Bohr's quantization of angular momentum,

$$L = \mu v r = n\hbar, \tag{5.12}$$

[10]This is found from a derivation analogous to the one leading to the gravitational result, Eq. (2.14). The zero of potential energy is taken to be zero at $r = \infty$.

to rewrite the kinetic energy, Eq. (5.10).

$$\frac{1}{8\pi\epsilon_0}\frac{e^2}{r} = \frac{1}{2}\mu v^2 = \frac{1}{2}\frac{(\mu v r)^2}{\mu r^2} = \frac{1}{2}\frac{(n\hbar)^2}{\mu r^2}.$$

Solving this equation for the radius r shows that the only values allowed by Bohr's quantization condition are

$$\boxed{r_n = \frac{4\pi\epsilon_0\hbar^2}{\mu e^2}n^2 = a_0 n^2,} \tag{5.13}$$

where $a_0 = 5.291772083 \times 10^{-11}$ m $= 0.0529$ nm is known as the **Bohr radius**. Thus the electron can orbit at a distance of $a_0, 4a_0, 9a_0, \ldots$ from the proton, but no other separations are allowed. According to Bohr's hypothesis, when the electron is in one of these orbits, the atom is stable and emits no radiation.

Inserting this expression for r into Eq. (5.11) reveals that the allowed energies of the Bohr atom are

$$\boxed{E_n = -\frac{\mu e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2} = -13.6 \text{ eV}\frac{1}{n^2}.} \tag{5.14}$$

The integer n, known as the **principal quantum number**, completely determines the characteristics of each orbit of the Bohr atom. Thus, when the electron is in the lowest orbit (the *ground state*), with $n = 1$ and $r_1 = a_0$, its energy is $E_1 = -13.6$ eV. With the electron in the ground state, it would take at least 13.6 eV to ionize the atom. When the electron is in the *first excited state*, with $n = 2$ and $r_2 = 4a_0$, its energy is greater than it is in the ground state: $E_2 = -13.6/4$ eV $= -3.40$ eV.

If the electron does not radiate in any of its allowed orbits, then what is the origin of the spectral lines observed for hydrogen? Bohr proposed that a photon is emitted or absorbed when an electron makes a transition from one orbit to another. Consider an electron as it "falls" from a higher orbit, n_{high}, to a lower orbit, n_{low}, without stopping at any intermediate orbit. (This is *not* a fall in the classical sense; the electron is *never* observed between the two orbits.) The electron loses energy $\Delta E = E_{\text{high}} - E_{\text{low}}$, and this energy is carried away from the atom by a single photon. Equation (5.14) leads to an expression for the wavelength of the emitted photon,

$$\boxed{E_{\text{photon}} = E_{\text{high}} - E_{\text{low}}}$$

or

$$\frac{hc}{\lambda} = \left(-\frac{\mu e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n_{\text{high}}^2}\right) - \left(-\frac{\mu e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n_{\text{low}}^2}\right),$$

which gives

$$\frac{1}{\lambda} = \frac{\mu e^4}{64\pi^3\epsilon_0^2\hbar^3 c}\left(\frac{1}{n_{\text{low}}^2} - \frac{1}{n_{\text{high}}^2}\right). \tag{5.15}$$

Comparing this with Eqs. (5.7) and (5.8) reveals that Eq. (5.15) is just the generalized Balmer formula for the spectral lines of hydrogen, with $n_{\text{low}} = 2$ for the Balmer series. Inserting values into the combination of constants in front of the parentheses shows that this term is exactly the Rydberg constant for hydrogen:

$$R_H = \frac{\mu e^4}{64\pi^3\epsilon_0^2\hbar^3 c} = 10967758.3\ \text{m}^{-1}.$$

This value is in perfect agreement with the experimental value quoted following Eq. (5.7) for the hydrogen lines determined by Johann Balmer, and this agreement illustrates the great success of Bohr's model of the hydrogen atom.[11]

Example 5.3.1. What is the wavelength of the photon emitted when an electron makes a transition from the $n = 3$ to the $n = 2$ orbit of the Bohr hydrogen atom? The energy lost by the electron is carried away by the photon, so

$$\begin{aligned} E_{\text{photon}} &= E_{\text{high}} - E_{\text{low}} \\ \frac{hc}{\lambda} &= -13.6\ \text{eV}\,\frac{1}{n_{\text{high}}^2} - \left(-13.6\ \text{eV}\,\frac{1}{n_{\text{low}}^2}\right) \\ &= -13.6\ \text{eV}\left(\frac{1}{3^2} - \frac{1}{2^2}\right). \end{aligned}$$

Solving for the wavelength gives $\lambda = 656.469$ nm in a vacuum. This result is within 0.03% of the measured value of the Hα spectral line, as quoted in Example 5.1.1 and Table 5.2.

The discrepancy between the calculated and the observed values is due to the measurements being made in air rather than in vacuum. Near sea level, the speed of light is slower than in vacuum by a factor of *approximately* 1.000297. Defining the **index of refraction** to be $n = c/v$, where v is the measured speed of light in the medium, $n_{\text{air}} = 1.000297$. Given that $\lambda\nu = v$ for wave propagation, and since ν cannot be altered in moving from one medium to another without resulting in unphysical discontinuities in the electromagnetic field of the light wave, the measured wavelength must be proportional to the wave speed. Thus $\lambda_{\text{air}}/\lambda_{\text{vacuum}} = v_{\text{air}}/c = 1/n_{\text{air}}$. Solving for the measured wavelength of the Hα line in air yields

$$\lambda_{\text{air}} = \lambda_{\text{vacuum}}/n_{\text{air}} = 656.469\ \text{nm}/1.000297 = 656.275\ \text{nm}.$$

continued

[11] The slightly different Rydberg constant, R_∞, found in many texts assumes an infinitely heavy nucleus. The reduced mass, μ, in the expression for R_H is replaced by the electron mass, m_e, in R_∞.

This result differs from the quoted value by only 0.0009%. The remainder of the discrepancy is due to the fact that the index of refraction is wavelength dependent. The index of refraction also depends on environmental conditions such as temperature, pressure, and humidity.[12]

Unless otherwise noted, throughout the remainder of this text, wavelengths will be assumed to be measured in air (from the ground).

The reverse process may also occur. If a photon has an energy equal to the *difference* in energy between two orbits (with the electron in the lower orbit), the photon may be absorbed by the atom. The electron uses the photon's energy to make an upward transition from the lower orbit to the higher orbit. The relation between the photon's wavelength and the quantum numbers of the two orbits is again given by Eq. (5.15).

After the quantum revolution, the physical processes responsible for Kirchhoff's laws (discussed in Section 5.1) finally became clear.

- A hot, dense gas or hot solid object produces a continuous spectrum with no dark spectral lines. This is the continuous spectrum of blackbody radiation emitted at any temperature above absolute zero and described by the Planck functions $B_\lambda(T)$ and $B_\nu(T)$. The wavelength $\lambda_{\max}$ at which the Planck function $B_\lambda(T)$ obtains its maximum value is given by Wien's displacement law, Eq. (3.15).

- A hot, diffuse gas produces bright emission lines. Emission lines are produced when an electron makes a downward transition from a higher orbit to a lower orbit. The energy lost by the electron is carried away by a single photon. For example, the hydrogen Balmer emission lines are produced by electrons "falling" from higher orbits down to the $n = 2$ orbit; see Fig. 5.6(a).

- A cool, diffuse gas in front of a source of a continuous spectrum produces dark absorption lines in the continuous spectrum. Absorption lines are produced when an electron makes a transition from a lower orbit to a higher orbit. If an incident photon in the continuous spectrum has exactly the right amount of energy, equal to

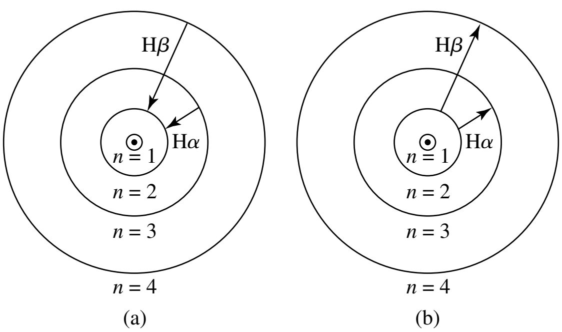

FIGURE 5.6 Balmer lines produced by the Bohr hydrogen atom. (a) Emission lines. (b) Absorption lines.

[12]See, for example, Lang, *Astrophysical Formulae*, 1999, page 185 for a fitting formula for $n(\lambda)$.

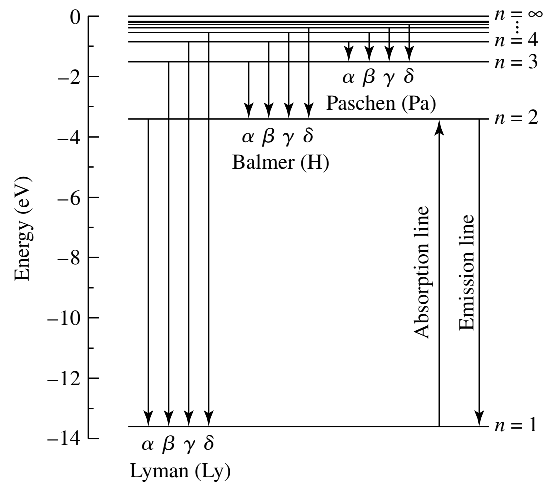

FIGURE 5.7 Energy level diagram for the hydrogen atom showing Lyman, Balmer, and Paschen lines (downward arrows indicate emission lines; upward arrow indicates absorption lines).

the difference in energy between a higher orbit and the electron's initial orbit, the photon is absorbed by the atom and the electron makes an upward transition to that higher orbit. For example, the hydrogen Balmer absorption lines are produced by atoms absorbing photons that cause electrons to make transitions from the $n = 2$ orbit to higher orbits; see Figs. 5.6(b) and 5.7.

Despite the spectacular successes of Bohr's model of the hydrogen atom, it is not quite correct. Although angular momentum is quantized, it *does not* have the values assigned by Bohr.[13] Bohr painted a *semiclassical* picture of the hydrogen atom, a miniature Solar System with an electron circling the proton in a classical circular orbit. In fact, the electron orbits are not circular. They are not even orbits at all, in the classical sense of an electron at a precise location moving with a precise velocity. Instead, on an atomic level, nature is "fuzzy," with an attendant uncertainty that cannot be avoided. It was fortunate that Bohr's model, with all of its faults, led to the correct values for the energies of the orbits and to a correct interpretation of the formation of spectral lines. This intuitive, easily imagined model of the atom is what most physicists and astronomers have in mind when they visualize atomic processes.

5.4 ■ QUANTUM MECHANICS AND WAVE–PARTICLE DUALITY

The last act of the quantum revolution began with the musings of a French prince, Louis de Broglie (1892–1987; see Fig. 5.8). Wondering about the recently discovered wave–particle duality for light, he posed a profound question: If light (classically thought to be a wave)

[13]As we will see in the next section, instead of $L = n\hbar$, the actual values of the orbital angular momentum are $L = \sqrt{\ell(\ell+1)}\,\hbar$, where ℓ, an integer, is a new quantum number.

FIGURE 5.8 Louis de Broglie (1892–1987). (Courtesy of AIP Niels Bohr Library.)

could exhibit the characteristics of particles, might not particles sometimes manifest the properties of waves?

de Broglie's Wavelength and Frequency

In his 1927 Ph.D. thesis, de Broglie extended the wave–particle duality to all of nature. Photons carry both energy E and momentum p, and these quantities are related to the frequency ν and wavelength λ of the light wave by Eq. (5.5):

$$\boxed{\begin{aligned} \nu &= \frac{E}{h} \\ \lambda &= \frac{h}{p}. \end{aligned}} \qquad \begin{aligned} (5.16) \\ (5.17) \end{aligned}$$

de Broglie proposed that these equations be used to define a frequency and a wavelength for *all* particles. The **de Broglie wavelength** and **frequency** describe not only massless photons but massive electrons, protons, neutrons, atoms, molecules, people, planets, stars, and galaxies as well. This seemingly outrageous proposal of matter waves has been confirmed in countless experiments. Figure 5.9 shows the interference pattern produced by *electrons* in a double-slit experiment. Just as Thomas Young's double-slit experiment established the wave properties of light, the electron double-slit experiment can be explained only by the wave-like behavior of electrons, with *each* electron propagating through *both* slits.[14] The wave–particle duality applies to everything in the physical world; everything exhibits its wave properties in its *propagation* and manifests its particle nature in its *interactions*.

[14]See Chapter 6 of Feynman (1965) for a fascinating description of the details and profound implications of the electron double-slit experiment.

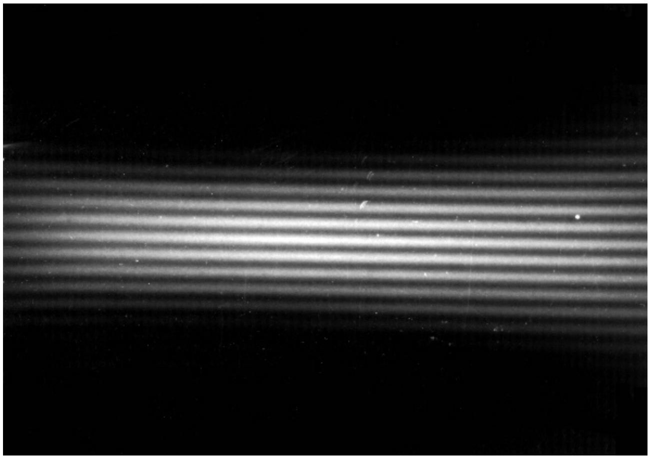

FIGURE 5.9 Interference pattern from an electron double-slit experiment. (Figure from Jönsson, *Zeitschrift für Physik*, *161*, 454, 1961.)

Example 5.4.1. Compare the wavelengths of a free electron moving at 3×10^6 m s^{-1} and a 70-kg man jogging at 3 m s^{-1}. For the electron,

$$\lambda = \frac{h}{p} = \frac{h}{m_e v} = 0.242 \text{ nm},$$

which is about the size of an atom and much shorter than the wavelength of visible light. Electron microscopes utilize electrons with wavelengths one million times shorter than visible wavelengths to obtain a much higher resolution than is possible with optical microscopes.

The wavelength of the jogging man is

$$\lambda = \frac{h}{p} = \frac{h}{m_{\text{man}} v} = 3.16 \times 10^{-36} \text{ m},$$

which is completely negligible on the scale of the everyday world, and even on atomic or nuclear scales. Thus the jogging gentleman need not worry about diffracting when returning home through his doorway!

Just what are the waves that are involved in the wave–particle duality of nature? In a double-slit experiment, each photon or electron must pass through *both* slits, since the interference pattern is produced by the constructive and destructive interference of the two waves. Thus the wave cannot convey information about where the photon or electron is, but only about where it *may* be. The wave is one of *probability*, and its amplitude is denoted by the Greek letter Ψ (psi). The square of the wave amplitude, $|\Psi|^2$, at a certain location describes the probability of finding the photon or electron at that location. In the double-slit experiment, photons or electrons are never found where the waves from slits 1 and 2 have destructively interfered—that is, where $|\Psi_1 + \Psi_2|^2 = 0$.

Heisenberg's Uncertainty Principle

The wave attributes of matter lead to some unexpected conclusions of paramount importance for the science of astronomy. For example, consider Fig. 5.10(a). The probability wave, Ψ, is a sine wave, with a precise wavelength λ. Thus the momentum $p = h/\lambda$ of the particle described by this wave is known exactly. However, because $|\Psi|^2$ consists of a number of equally high peaks extending out to $x = \pm\infty$, the particle's location is perfectly uncertain. The particle's position can be narrowed down if several sine waves with different wavelengths are added together, so they destructively interfere with one another nearly everywhere. Figure 5.10(b) shows the resulting combination of waves, Ψ, is approximately zero everywhere except at one location. Now the particle's position may be determined with a greater certainty because $|\Psi|^2$ is large only for a narrow range of values of x. However, the value of the particle's momentum has become more uncertain because Ψ is now a combination of waves of various wavelengths. This is nature's intrinsic trade-off: The uncertainty in a particle's position, Δx, and the uncertainty in its momentum, Δp, are inversely related. As one decreases, the other must increase. This fundamental inability of a particle to *simultaneously* have a well-defined position and a well-defined momentum is a direct result of the wave–particle duality of nature. A German physicist, Werner Heisenberg (1901–1976), placed this inherent "fuzziness" of the physical world in a firm theoretical framework. He demonstrated that the uncertainty in a particle's position multiplied by the uncertainty in its momentum must be *at least* as large as $\hbar/2$:

$$\boxed{\Delta x\, \Delta p \geq \frac{1}{2}\hbar.} \tag{5.18}$$

Today this is known as **Heisenberg's uncertainty principle**. The equality is rarely realized in nature, and the form often employed for making estimates is

$$\boxed{\Delta x\, \Delta p \approx \hbar.} \tag{5.19}$$

A similar statement relates the uncertainty of an energy measurement, ΔE, and the time interval, Δt, over which the energy measurement is taken:

$$\boxed{\Delta E\, \Delta t \approx \hbar.} \tag{5.20}$$

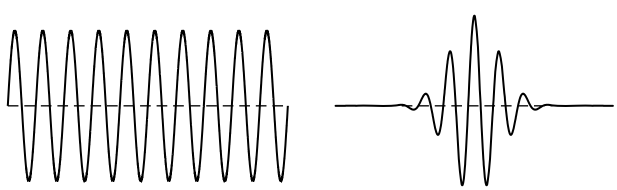

FIGURE 5.10 Two examples of a probability wave, Ψ: (a) a single sine wave and (b) a pulse composed of many sine waves.